

The Gongfu of Xinzuo

A Practical Guide to Kitchen Knives - Editorial review edition

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Contents

1	Preface	1
2	The Gongfu of Xinzuo	3
3	Anatomy of a Kitchen Knife	5
3.1	Read the tool from edge to hand	5
3.2	Blade terminology	6
3.3	Handle, bolster and tang	6
3.4	What this book calls an “Asian handle”	7
3.5	Hardness, edge retention and flexibility	7
3.6	Read the blade face through three separate questions	8
4	Knife Shapes and Their Uses	11
4.1	General-purpose knives	12
4.1.1	Western chef’s knife	12
4.1.2	Gyuto	13
4.1.3	Santoku	13
4.1.4	Bunka	14
4.2	Vegetable and Chinese all-purpose knives	15
4.2.1	Nakiri	15
4.2.2	Chinese cleaver	16
4.2.3	Bone chopper	17
4.3	Long carving and slicing knives	18
4.3.1	Classic carving knife	18
4.3.2	Roast carving knife	18
4.3.3	Ham knife	19
4.3.4	Sashimi knife	19
4.3.5	Sakimaru	19
4.3.6	Kiritsuke	21
4.4	Boning, filleting and fish butchery	21

4.4.1	Western boning knife	21
4.4.2	Honesuki	22
4.4.3	Fillet knife	23
4.4.4	Deba	23
4.5	Small, table and precision knives	24
4.5.1	Utility knife	24
4.5.2	Curved paring knife	25
4.5.3	Straight paring knife	25
4.5.4	Steak knife	25
4.5.5	Cheese knife	26
4.6	Serrated and hard-food specialists	27
4.6.1	Bread knife	27
4.6.2	Frozen-food knife	28
4.7	Choosing by work rather than by name	28
5	Weight, Balance and Ergonomics	31
5.1	Weight changes the gesture	31
5.2	Where the weight sits changes the knife	32
5.3	Balance depends on the grip	33
5.4	Ergonomics is part of performance	34
5.5	How staff should guide the choice	34
6	Single and Double Bevels	37
6.1	Double bevel: the familiar all-purpose edge	38
6.1.1	How it feels during a cut	38
6.2	Not every double bevel is symmetrical	39
6.3	Single bevel: one side does most of the shaping	39
6.3.1	The main bevel may change angle from heel to tip	39
6.4	The urasuki is not an unfinished back	40
6.5	Handedness and working direction	40
6.6	How steering develops	41
6.7	Clean surfaces and food separation	41
6.8	Strength does not come from bevel count alone	42
6.9	Choosing between the two	42
6.10	What the geometry means for sharpening	43
6.11	What to remember	43
7	How to Cut - Essential Knife Techniques	45
7.1	Before the first cut	45
7.2	Read the path before naming the technique	46
7.3	Board contact: direct descent	47
7.3.1	What the movement looks like	47
7.3.2	Try it with a cucumber	47

7.3.3	Why it works	47
7.3.4	Common mistake	47
7.4	Board contact: advancing stroke	47
7.4.1	What the movement looks like	47
7.4.2	Try it with a carrot	48
7.4.3	Why it works	48
7.4.4	A memory aid	48
7.5	Long work: draw through the food	48
7.5.1	What the movement looks like	48
7.5.2	Try it with a boneless cooked chicken breast	49
7.5.3	Why it works	49
7.6	Pivoted work: keep a point of contact	49
7.6.1	Wide rolling rock	49
7.6.2	Why the round-tip nakiri suits a compact rock	49
7.6.3	Try it with parsley	50
7.6.4	Common mistake	50
7.7	Tip work	50
7.8	Slicing across the grain	51
7.9	Fine slicing and julienne	51
7.10	Dicing	51
7.11	Mincing without crushing	52
7.12	Crushing and transferring food	52
7.13	Peeling and in-hand work	52
7.14	Working around bones and joints	53
7.15	Filleting fish	53
7.16	Specialist techniques that need demonstration	53
7.17	A seven-day practice sequence	54
7.18	What to remember	54
8	The Five Dimensions of Knife Steel	55
8.1	1. Hardness: resistance to permanent deformation	56
8.2	2. Sharpening response: more than “soft equals easy”	56
8.3	3. Corrosion resistance: what “stainless” means	57
8.4	4. Toughness: resistance to fracture	58
8.5	5. Microstructural fineness: what supports the apex	58
8.6	The five dimensions interact	59
8.7	A worked example: reading the Xinzuo range	59
9	Alloying Elements	61
9.1	Follow each element through the steel	61
9.2	Carbon: what makes hard knife steel possible	63
9.3	Chromium: the basis of stainless behaviour	64
9.4	Molybdenum: reliable hardening and pitting resistance	65

9.5	Vanadium: fine structure and very hard carbides	65
9.6	Tungsten: durable carbides and high-temperature stability . . .	66
9.7	Cobalt: supporting demanding heat treatment	66
9.8	Nickel: toughness and visual contrast	66
9.9	Nitrogen: another route to hardness and corrosion resistance .	67
9.10	Why percentages are not enough	67
10	Conventional and Powder-Metallurgy Steels	69
10.1	First, picture the inside of steel	69
10.2	Why a large ingot can become uneven	70
10.3	Powder metallurgy: solidifying millions of tiny droplets	70
10.4	Why this matters at a knife edge	71
11	Heat Treatment and Carbides: Turning Composition into a Blade	73
11.1	What steel looks like under a microscope	73
11.2	Matrix, grains and hard particles	74
11.3	What carbides are	74
11.4	1. Preparing the steel before hardening	75
11.5	2. Austenitizing: creating the high-temperature phase	75
11.5.1	Vacuum heat treatment: controlling the atmosphere . .	76
11.6	3. Quenching: transforming austenite into martensite	76
11.7	4. Retained austenite and sub-zero treatment	77
11.7.1	Xinzuo as an example	77
11.8	5. Tempering: making martensite usable	78
11.9	6. Grinding after heat treatment	78
11.10	The main carbide families in knife steels	78
11.11	Why carbide hardness changes sharpening	80
11.12	Why carbide size and amount affect toughness	80
11.13	Why heat treatment changes the carbide population	80
12	Xinzuo Blade Steels: Different Balances for Different Knives	83
12.1	A compact comparison	83
12.2	Premium powder-metallurgy steels	91
12.2.1	ZDP-189: maximum hardness and long edge life	91
12.2.2	SRS13: premium performance without ZDP-189's extreme hardness	91
12.2.3	Böhler M390: wear and corrosion resistance first	91
12.2.4	14Cr14MoVNB: Xinzuo's niobium-bearing premium powder steel	92
12.3	Carbon and all-round high-performance stainless steels	92
12.3.1	Blue Steel No. 2: high hardness, responsive sharpening .	92

12.3.2	10Cr15CoMoV: the long-standing Xinzuo all-round reference	93
12.3.3	12Cr18MoV: the newer everyday-use formulation	93
12.3.4	AUS-10: a Japanese all-rounder near the VG10 class	94
12.3.5	VG10: the Japanese benchmark and why it matters to Xinzuo	94
12.3.6	Sandvik 14C28N: fine structure, nitrogen and edge stability	95
12.4	More forgiving stainless steels	95
12.4.1	440C: the classic high-carbon stainless work steel	95
12.4.2	DIN 1.4116: where its forgiveness comes from	96
12.5	Layered constructions are not separate “sharpness levels”	96
12.6	Reading the Xinzuo steel range	97
13	Xinzuo Handle Materials - Feel, Care and Choice	99
13.1	Read the material by design consequence	100
13.2	Synthetic and technical composite materials	101
13.2.1	Polypropylene (PP)	101
13.2.2	ABS	101
13.2.3	G10	102
13.2.4	Carbon fibre	102
13.2.5	Carbon fibre with resin	102
13.2.6	Resin	103
13.3	Semi-synthetic and hybrid materials	103
13.3.1	Micarta	103
13.3.2	Pakka wood	104
13.3.3	Stabilized wood	104
13.4	Natural materials	105
13.4.1	Rosewood	105
13.4.2	Ebony	105
13.4.3	North American desert ironwood	105
13.4.4	Red sandalwood and red wood	106
13.4.5	Figured sycamore	106
13.4.6	Olivewood	106
13.4.7	Oak and burnt oak	107
13.4.8	Walnut	107
13.5	Accent and combination materials	107
13.6	Material, ergonomics and balance	107
13.7	What a material name does not establish	108
13.8	Choosing a handle by use	108
13.9	Care by failure mode	109
14	Damascus Steel - Pattern, History and Construction	111
14.1	Start at the cutting edge	112

14.2	How a welded pattern becomes visible	112
14.3	What layer count can tell us	114
14.4	Performance follows construction	114
14.5	Surface decoration is a separate category	115
14.6	Historical crucible steel is a different route	115
14.7	Bintie and the Chinese part of the story	116
14.8	From the historical question to Xinzuo	116
14.9	A reliable reading sequence	117
15	The Self-Sharpening Effect in Full Damascus Blades	119
15.1	Why full Damascus matters	119
15.2	The role of different steels	120
15.3	What happens during use	120
15.4	Sharpness and cutting aggression are different	120
15.5	What self-sharpening does not mean	121
15.6	Comparing it with a serrated knife	121
15.7	Why use still matters	121
15.8	Sharpening a self-sharpening full Damascus blade	122
15.9	What this means for a Xinzuo knife	122
16	Routine Knife Care: What to Do Between Sharpenings	123
16.1	After every use: wash, rinse and dry	123
16.2	Why dishwashers are poor for quality knives	124
16.3	Stainless steel still needs care	124
16.4	Carbon steel, patina and rust	124
16.5	The cutting board is part of edge maintenance	125
16.6	Do not scrape the board with the cutting edge	125
16.7	Storage: protect the edge from hard objects	125
16.8	Natural-wood handles	126
16.9	Composite handles	126
16.10	What about oiling the blade?	126
16.11	Honing is not sharpening—and rods do different things	126
16.12	Inspect before damage becomes serious	127
16.13	Routine care is mostly prevention	127
17	Safe Use and Carrying	129
17.1	Create a safe working area before cutting	129
17.2	Stabilize the ingredient	130
17.3	Use the right knife for the task	130
17.4	Grip and control	131
17.5	Let the edge do the work	131
17.6	Use a suitable cutting board	132
17.7	Avoid twisting and sideways scraping	132

17.8 Wash safely as well as carefully	132
17.9 Passing a knife to another person	133
17.10 Moving with a knife in the kitchen	133
17.11 Transport beyond the workstation	133
17.12 Safe storage	134
17.13 What to do if a knife falls	134
17.14 Sharpness and safety	134
17.15 Common unsafe habits	135
17.15.1 Leaving a knife in the sink	135
17.15.2 Cutting toward the body	135
17.15.3 Holding food in the air with a large knife	135
17.15.4 Testing sharpness by sliding a finger along the edge . . .	135
17.15.5 Scraping the board with the cutting edge	135
17.15.6 Using a glass, ceramic, stone or metal board	135
17.15.7 Trying to catch a falling knife	135
17.16 What to remember	136

18 Your First Successful Hand Sharpening 137

18.1 What you are actually sharpening	138
18.2 What you need	138
18.3 Inspect the knife before sharpening	139
18.4 Prepare the stone and your workspace	139
18.5 Find the microbevel with a marker	140
18.6 Learn one complete-edge pass	141
18.6.1 How much pressure to use	142
18.7 The complete beginner method	142
18.7.1 Step 1: make the starting map	142
18.7.2 Step 2: mark the first microbevel	142
18.7.3 Step 3: sharpen the first side	143
18.7.4 Step 4: sharpen the second side	143
18.7.5 Step 5: reduce and remove the burr	143
18.7.6 Step 6: use a finer stone only after the medium stone succeeds	144
18.7.7 Step 7: clean before testing	144
18.8 Understand and check the burr safely	144
18.9 Three checks after sharpening	145
18.9.1 Check 1: reflected light	145
18.9.2 Check 2: paper in three sections	145
18.9.3 Check 3: food and stability	145
18.10 Workshop examples	146
18.10.1 The middle cuts, but the tip catches paper	146
18.10.2 The knife cuts paper once, then feels dull	146

18.10.3	The marker disappears above the microbevel but remains at the edge	146
18.10.4	The marker disappears only from the very edge	146
18.10.5	The burr appears at the heel and middle but not at the tip	147
18.10.6	One side forms a burr quickly and the other takes longer	147
18.10.7	The blade sticks and the movement becomes jerky . . .	147
18.10.8	The knife cuts paper but wedges in a carrot	147
18.11	Mistakes that matter most	147
18.11.1	Sharpening the primary bevel instead of the microbevel	147
18.11.2	Changing the angle during every pass	147
18.11.3	Pressing harder when nothing happens	148
18.11.4	Stopping the pass in the middle	148
18.11.5	Chasing a large burr	148
18.11.6	Testing only the middle	148
18.11.7	Starting on a damaged or expensive knife	148
18.12	Base-level exercises	148
18.12.1	Exercise 1: stable setup	148
18.12.2	Exercise 2: the marker line	148
18.12.3	Exercise 3: the burr map	149
18.12.4	Exercise 4: pressure reduction	149
18.12.5	Exercise 5: the three-part paper test	149
18.12.6	Exercise 6: edge stability	149
18.12.7	Exercise 7: repeat the result	149
18.13	Your base-level test	149
19	Improving Your Sharpening Skills	151
19.1	The three skills behind better sharpening	151
19.2	Read the evidence after every stage	152
19.2.1	Practice drill: predict before looking	152
19.3	Build three pressure levels	152
19.3.1	Working pressure	152
19.3.2	Refining pressure	153
19.3.3	Finishing pressure	153
19.3.4	Practice drill: make the burr smaller	153
19.4	Introduce section work carefully	153
19.4.1	Workshop example: one dull centimetre near the heel .	154
19.4.2	Section-work test	154
19.5	Choose stones by the work	154
19.5.1	Hard and soft stones	155
19.5.2	Water, slurry and a readable surface	155
19.6	Keep the stone flat	155
19.6.1	Flatness test	155
19.7	Distinguish apex work from geometry work	156

19.7.1	A practical geometry comparison	156
19.7.2	Recognise when thinning may help	156
19.7.3	Geometry test	157
19.8	Choose the finish for the food	157
19.9	Improve deburring before adding polish	157
19.9.1	Stability test	157
19.10	Approach asymmetry with evidence	158
19.11	Traditional single-bevel knives	158
19.11.1	Hamaguri in plain language	159
19.12	A five-session improvement plan	159
19.12.1	Session 1: contact	159
19.12.2	Session 2: pressure	159
19.12.3	Session 3: burr control	159
19.12.4	Session 4: local correction	159
19.12.5	Session 5: task finish	159
19.13	The second-level test	160
19.14	What expert practice adds	160

20 Bibliography **161**

20.1	The Five Dimensions of Knife Steel	161
20.2	Alloying Elements	161
20.3	Conventional and Powder-Metallurgy Steels	162
20.4	Heat Treatment and Carbides	162
20.5	Xinzuo Blade Steels	162
20.6	Xinzuo Handle Materials - Feel, Care and Choice	163
20.7	Damascus Steel - Pattern, History and Construction	163
20.8	Knife Shapes and Their Uses	163
20.9	Weight, Balance and Ergonomics	163
20.10	Learning Hand Sharpening	164

Chapter 1

Preface

The title of this book comes from an idea much broader than martial arts. *Gongfu* can describe a skill built through time, study, repetition and disciplined practice: knowledge turned into reliable ability.

That is the spirit of this book. A kitchen knife looks simple, but using one well takes more than learning a few movements. Skill grows faster when practice is supported by understanding: when the user knows why a knife behaves as it does, can separate meaningful differences from superficial ones and can connect what is felt in the hand with the choices behind the tool.

This book provides a technical and practical foundation for developing that competence. It cannot replace years of professional experience, and reading alone cannot create mastery. Its purpose is to make practice more effective: to help the reader observe carefully, understand established solutions, ask better questions and learn faster from experience. *Gongfu* begins by learning what others have discovered, then practising until those principles can be understood, judged, adapted and eventually improved.

The book combines personal study and experience with knowledge developed inside Xinzuo through design, manufacturing, testing, sales and daily contact with cooks and customers. It brings that knowledge together to help readers understand Xinzuo knives and develop the judgement needed to choose, use and care for them well.

Knowledge improves when it is shared. This book is also available free of charge on GitHub at <https://github.com/francescomazza/xinzuo-kung-fu>. The repository supports collaborative revision, and corrections,

comments and proposals are welcome there or by email at **francescoclaudio-mazza@gmail.com**.

This book is a starting point: knowledge becomes practice, practice becomes experience, and experience can become *gongfu*.

Francesco Claudio Mazza *EU Brand and Operations Manager, Xinzuo*

Chapter 2

The Gongfu of Xinzuo

A Practical Guide to Kitchen Knives Materials, Shapes and Techniques for Masterful Cutting

A kitchen knife looks simple: a handle, a blade and a very thin edge. In use, it can chop, glide, rock, steer, wedge, separate or crush. What the cook feels comes from the interaction between the knife, the ingredient and the cutting movement.

A kitchen knife combines material, process, geometry and craftsmanship.

This book explains those interactions and builds practical skill in choosing, using and caring for kitchen knives. Technical concepts appear when they explain something that can be seen or felt in use, so knowledge can grow into judgement and skill.

Xinzuo products provide concrete examples from real design, manufacturing and use. The goal is to understand why differences between knives matter in the hand and on the cutting board, not to memorize product names or specifications.

The central idea: steel creates possibilities, heat treatment develops them, geometry turns them into cutting behaviour, sharpening restores the edge and technique determines what the cook experiences.

The Preface explains the meaning of *gongfu* in the title, the purpose of the book and the background behind it.

Chapter 3

Anatomy of a Kitchen Knife

Before comparing blade shapes, steels or sharpening angles, read a knife as three connected systems: the cutting geometry, the steel structure that carries it and the hand interface that controls it. The names below locate each feature within those systems.

Not every knife has every feature shown here. A Western full-tang chef's knife, a molded-handle utility knife and an Asian-handle knife can all be excellent tools with very different constructions.

3.1 Read the tool from edge to hand

A fixed kitchen knife begins with the **blade**. Its front ends at the **tip**; the upper unsharpened side is the **spine**; the broad side is the **blade face**; the sharpened lower boundary is the **cutting edge**; and the rear of the cutting portion is the **heel**. The sloping geometry leading to the edge is the **bevel**. The visible texture or polish on the blade face is the **surface finish**.

Behind the blade, the steel continues as the **tang**. On many Western knives the tang is visible around the handle and covered on both sides by **handle scales**, usually fixed with adhesive and **rivets or pins**. Many Western kitchen knives also have a metal **bolster** between blade and handle. The rear end of the handle is the **butt** or **pommel**.

Kitchen knife anatomy

Two labelled diagrams. The upper diagram shows a Western full-tang kitchen knife with the cutting edge facing downward. The lower diagram shows the hidden-tang octagonal construction called an Asian handle in this book.

Western full-tang construction Side view — spine above, cutting edge below

Butt / pommel Handle scale Bolster Spine Tip

Full tang Rivet / pin Heel Bevel Blade face Cutting edge surface finish appears on this face

Asian handle Octagonal handle with ferrule and hidden tang

octagonal cross-section

Handle Ferrule Hidden tang Blade

The upper diagram shows a common Western full-tang construction. The lower diagram shows the “Asian handle” used in this book: an octagonal handle, front ferrule and hidden tang.

3.2 Blade terminology

The **tip** is the front of the blade. Its shape affects how easily the knife enters small spaces, scores a surface or makes fine directional corrections. The **spine** is the upper, normally unsharpened side. Its thickness and taper toward the tip affect stiffness, balance and how easily the blade passes through food.

The broad side is the **blade face**. It is not always flat: convexity, hollows, cladding transitions and other shaping can change how food moves against it. The **surface finish** is the final texture or appearance on that face, such as satin, polished, mirror-polished, brushed, etched or hammered. Finish and geometry are different: two blades can look similar while having very different thickness and cross-section.

The **cutting edge** or **edge** is the sharpened line that enters the food. Above it, the **bevel** narrows the blade toward the apex. Some knives have a clearly visible primary bevel; on others the transition is subtle. The **heel** is the rear of the cutting edge, closest to the handle. Depending on the design, a short unsharpened area or relief may separate it from the bolster or handle.

3.3 Handle, bolster and tang

The **handle** is the part held by the user, but its construction varies. On a Western **full-tang** knife, the tang runs through almost the full length and

outline of the handle. **Scales** cover its sides and are usually secured with adhesive and **rivets or pins**. The exposed steel can add mass and affect balance.

A **bolster** is the reinforced metal transition between blade and handle found on many Western kitchen knives. It may be integral with the blade or attached separately. It can strengthen the transition, create a comfortable finger contact point, protect the front of the handle and add mass near the hand. Some bolsters extend to the heel; others leave the edge open for easier sharpening. A bolster is not necessarily a **guard**. A guard is specifically designed to stop the hand sliding toward the edge, and many kitchen knives have no separate guard.

Full tang is only one valid construction. A **hidden tang** is narrower than the handle and enclosed inside it. A **through tang** reaches the butt and may be mechanically secured there. A **partial or stub tang** ends before the butt. Molded handles can encapsulate the tang so no scales or rivets are visible. None of these constructions creates a simple quality hierarchy. Durability depends on tang dimensions, materials, fit, fastening, intended use and expected loads.

3.4 What this book calls an “Asian handle”

In this book, **Asian handle** means the traditional hidden-tang construction commonly paired with an **octagonal**, and sometimes oval, handle. The blade enters a solid handle on a hidden tang, while a **ferrule** or collar often reinforces the front and forms the transition to the blade.

The term is intentionally narrow. It does **not** mean that all knives made in Asia use this construction; many use Western full-tang, molded or other systems. Here, **Asian handle** means the octagonal-or-oval hidden-tang construction with a front ferrule described above.

A ferrule is also different from a Western bolster. Both sit near the blade-handle junction, but the ferrule is mainly a collar around the front of the handle material, while the bolster is a metal component at the blade-handle transition.

3.5 Hardness, edge retention and flexibility

For knife selection, remember: **geometry determines how easily a blade bends; steel and heat treatment determine how safely it tolerates that bending.**

A thin blade flexes more easily than a thick one, but a knife designed to flex repeatedly also needs enough toughness to survive lateral stress. With comparable geometry and broadly similar steel families, **higher hardness supports a fine edge and usually improves edge retention, but reduces tolerance for severe bending, twisting and impact.** Forced too far sideways, a very hard blade is more likely to chip, crack or break than simply remain permanently bent.

Many high-end cutting knives use hard steels and relatively rigid precision geometries. They keep a fine edge for a long time but should not be tested by bending them. Knives designed to strike bone or flex around contours usually use tougher, more forgiving steels or lower-hardness heat treatments. They tolerate impact and deflection better but normally need more frequent edge maintenance.

A useful memory aid is:

harder precision blade = longer edge retention, less tolerance for forced flex

tougher/softer working blade = more tolerance for flex or impact, less edge retention

This is a training rule, not a complete metallurgical formula. Blade thickness, taper, composition, the steel's microscopic internal structure, heat treatment and edge geometry still matter. The rule explains why a flexible fillet knife, a bone chopper and a very hard fine-edged slicer use different compromises.

3.6 Read the blade face through three separate questions

Do not treat every visible surface feature as the same kind of treatment.

First: what is the topography? Hollows, channels and hammered depressions change the physical shape of the blade face. By interrupting broad contact, they may reduce suction or sticking with some moist or starchy foods. Depth, position and spacing matter more than the decorative name. A chef's knife with hollows remains a chef's knife because its working profile has not changed category.

Second: what texture did finishing leave? Satin grinding leaves directional lines; polishing reduces surface roughness and a mirror finish carries that process further. Texture affects drag, scratch visibility and cleaning, but it does not create sharpness at the apex.

Third: did the surface reveal or retain a structure? Etching can expose contrast between Damascus layers. Hand forging can leave irregular scale or texture when the face is not ground completely smooth. A retained forged surface and deliberate hammering may look related, but they come from different processes and should be described separately.

Food release depends on topography, blade geometry, finish, food and movement together. No pattern or polish can replace suitable thickness and cross-section.

With these general principles established, we can compare blade shapes without repeating them in every profile.

Chapter 4

Knife Shapes and Their Uses

Knife shape connects the cook directly to the work. Steel, heat treatment and construction affect how the edge behaves; the blade outline affects how the hand approaches food. Length sets the useful stroke. Height creates knuckle clearance. Curvature favours some movements over others. Thickness affects cutting resistance and how much abuse the blade can tolerate. The point may be a precision tool, a safe rounded end or almost absent.

This chapter is **not an exhaustive catalogue of every knife shape**. It covers the forms most common in the Xinzuo range and everyday kitchen work, plus a few useful specialist profiles. Many other regional, historical and specialist shapes exist, but including all of them would make the chapter harder to use.

For sales staff, one principle matters more than memorising names: **start from the customer's work, not from the silhouette they like**. Ask what foods they cut most often, in what quantities, whether they work mainly on a board or in the hand, and whether they need broad chopping, long slicing, work around bone, fish preparation or fine detail. These answers identify the blade family that best suits the job.

After choosing the right shape, move to ergonomics. Ask the customer to hold the knife as they actually cook and compare handle shape, grip position, total weight and balance. Some people want a planted, blade-forward knife that feels authoritative in a push cut; others prefer a lighter or more neutral knife

that is easy to lift and redirect. The recommendation is complete only when the functional shape is matched to the handle and balance that suit the customer's hand and movement.

Xinzuo's broad range is especially useful at this stage. Staff can put several technically suitable knives in the customer's hand and let them compare. This makes it possible to find a close ergonomic fit instead of treating one handle or balance as universal. Use a clear order: **function first, ergonomics second, specific model third.**

Read the actual blade from heel to tip and ask what its geometry encourages the hand to do. Category names are only guides. Two knives sold under the same name can differ greatly in curve, thickness, balance, bevel geometry and intended motion. No steel grade, finish or specification can make the wrong shape right for the job.

A flatter heel and middle give broad board contact and favour push cutting and nearly vertical chopping. A continuous belly lets the contact point travel along the edge during drawing or rocking. A long, narrow blade keeps more edge in contact during a draw slice and reduces drag from the blade face. A narrow point reaches joints and fine details; a broad, supported heel tolerates controlled force better. These are tendencies, not rigid rules: a gyuto can have enough curve to rock, a Western chef's knife can be fairly flat, and a rectangular Chinese-style knife may be a thin general-purpose cutter rather than a bone tool.

With these relationships in mind, the main knife families are easier to understand.

4.1 General-purpose knives

4.1.1 Western chef's knife

The Western chef's knife is a versatile general-purpose form with a continuous belly curve. The curve makes sweeping, drawing and rocking natural, especially when resistance changes through a cut in meat, connective tissue or mixed ingredients. As the knife moves, the working point travels along the belly, combining forward motion with penetration instead of relying only on vertical pressure.

It also handles vegetables and fruit well, which is why it became a familiar all-round kitchen tool. Its broad, often substantial blade is usually harder to redirect than a thin narrow slicer once deeply engaged in food. It is a fine



Figure 4.1: A Xinzuo B46D Western chef’s-knife example showing the characteristic curved belly.

cutting tool, not a bone chopper: hard bone, frozen food, twisting and levering are outside its intended use.

4.1.2 Gyuto

IMAGE PLACEHOLDER

Xinzuo PM80-CS gyuto profile</div>

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The gyuto overlaps strongly with the Western chef’s knife but often has a flatter heel and middle, a slimmer point and less continuous belly. It tends to feel more linear in push cuts and long draw slices and can be easier to steer through small directional changes.

Less curvature usually means less progressive rocking than a strongly bellied Western chef’s knife, while the slimmer front improves precision. This does not make the gyuto unsuitable for meat or the Western chef’s knife unsuitable for vegetables. The difference is one of emphasis. Actual examples vary, so curve, height, thickness and balance matter more than the label.

4.1.3 Santoku

The santoku is a compact, **primarily vegetable-oriented but highly versatile** knife. Its tall blade, relatively straight heel and middle and lowered



Figure 4.2: A Xinzuo santoku example: compact in length but generous in blade height.

front give broad board contact for push cutting and nearly vertical chopping, while enough curve remains for controlled slicing. The height provides knuckle clearance and a useful guiding surface, making it especially comfortable for vegetables, fruit and herbs while still handling boneless meat and fish well.

A santoku usually keeps more steel around the front than a fine pointed chef's knife or bunka. This fuller nose gives up some point precision for stronger support and a more planted feel. The hand naturally prepares a firmer stroke when controlling the extra front mass, and the moving blade carries more momentum through resistance. **All else equal, this gives the santoku more authority in practical push cutting and short vertical chopping.**

Its main limitation is compactness: a short edge needs more strokes across a large roast, melon or other wide ingredient, and a very flat santoku is not designed for exaggerated rocking.

4.1.4 Bunka

The bunka shares the santoku's compact height and board-friendly versatility but ends in an angular **K-tip**. The important feature is not simply a pointed nose: the K-tip places a fine working point low and close to the cutting line, where it is easy to see and control.

This combines **the working surface and knuckle clearance of a relatively large blade with front-end precision close to that of a utility knife**. The tip can score skin, enter a small seam, trim delicate areas, make shallow



Figure 4.3: A Xinzuo bunka example with its characteristic angular K-tip.

incisions and perform fine directional corrections without sacrificing the broad blade needed for normal board work.

The form is especially well suited to **vegetables and fish**, including fine vegetable preparation, trimming, portioning and detail work. It also handles many boneless-meat tasks. The fine K-tip tolerates twisting, levering and hard impact less well than the fuller nose of a santoku; that is the price of added precision.

4.2 Vegetable and Chinese all-purpose knives

4.2.1 Nakiri

The nakiri is a thin, tall, double-bevel vegetable knife built for broad board contact. Its relatively straight edge makes push cutting and vertical chopping efficient, while its height gives room for the guiding hand and for transferring prepared vegetables.

Nakiri fronts vary. Some have a strongly rounded, unsharpened front corner; others carry the sharpened edge farther forward. A round-tip nakiri works especially well for compact rock chopping and fine mincing. The rounded front can pivot on the board, the tall blade gives knuckle clearance and a clear guide against the non-cutting hand, and the straight working edge keeps useful board contact through a short controlled arc. This creates a compact rocking motion without the deep belly of a Western chef's knife. It works particularly well for herbs and finely chopped vegetables, and the rounded front is less vulnerable



Figure 4.4: A Xinzuo nakiri example: tall, thin and nearly straight along the cutting edge.

than a fine K-tip during repeated board work.

The nakiri is still a thin vegetable cutter and is not intended for bone, frozen food or impact.

4.2.2 Chinese cleaver



Figure 4.5: A Xinzuo Chinese-cleaver example; the exact specification determines whether it is a fine cutter or a heavier tool.

The Chinese cleaver is one of the most distinctive all-purpose forms. Its tall rectangular blade can slice, chop, guide against the knuckles, crush aromat-

ics and transfer prepared ingredients. Examples vary greatly in weight and thickness, so the silhouette alone never proves that a knife can cut bone.

Its height, forward mass and lack of a familiar pointed profile can make it harder for beginners because they change the usual visual and tactile references for fine control. In experienced hands, the same features make it exceptionally versatile. Always read weight, thickness behind the edge, bevel geometry and intended use.

4.2.3 Bone chopper



Figure 4.6: A Xinzuo bone-chopper example, built around mass and edge support rather than delicate slicing.

A bone chopper uses mass, a supported edge and substantial cross-section for controlled butchery where durability and momentum matter more than low-drag slicing. Steels and heat treatments for this work usually favour toughness over extreme hardness because the blade must absorb impact without failing suddenly.

It is not indestructible. Large dense bones, hard frozen material, unstable work or a glancing strike can damage the knife and injure the user. A thin Chinese cleaver may look similar while having completely different capabilities.



Figure 4.7: A Xinzuo classic carving-knife example with a long, narrow profile.

4.3 Long carving and slicing knives

4.3.1 Classic carving knife

A carving knife is longer and narrower than a chef's knife so it can cross a roast, poultry breast, boneless meat or large fruit with fewer strokes. Less blade height reduces drag and improves visibility. The intended movement is a long draw, not repeated short sawing.

4.3.2 Roast carving knife



Figure 4.8: A Xinzuo roast-carving example with a broader, upswept front.

The roast-carving profile keeps a long slicing edge but gives the front more breadth and upward sweep. The curve follows the rounded surface of a roast and lets a draw slice finish with a smooth rising motion. It is still a slicer for cooked, boneless food, not a joint breaker.

4.3.3 Ham knife



Figure 4.9: A Xinzuo ham-knife example, extra long and narrow for broad, thin slices.

A ham knife extends the carving-knife concept. Its very long, narrow blade produces broad thin slices with little friction. Some examples flex slightly to follow the ham; others are firmer. The design excels at long slicing but is inefficient for chopping and small board work.

4.3.4 Sashimi knife

Long traditional sashimi slicers use their length to cut raw fish in one pulling stroke, leaving a cleaner surface than repeated back-and-forth motion. Traditional examples are single bevelled, normally handed and sharpened differently from ordinary double-bevel slicers. This specialised geometry can steer if the user does not understand the bevel.

4.3.5 Sakimaru

The sakimaru belongs to the same long single-bevel slicing family but ends in a rounded, sword-influenced front. This changes balance, presentation and front access without changing the main task: clean draw slicing through boneless fish.



Figure 4.10: A Xinzuo long single-bevel sashimi slicer.



Figure 4.11: A Xinzuo sakimaru example with its rounded, sword-influenced point.

4.3.6 Kiritsuke



Figure 4.12: A Xinzuo traditional single-bevel kiritsuke example with a long edge and angular K-tip.

In the specialist form shown here, kiritsuke means a long, angular-tipped single-bevel knife for precise slicing and advanced preparation. The K-tip enters delicate cuts accurately. The knife can also be controlled with the index finger extended along the spine toward the front, placing tactile guidance close to the point for shallow incisions and fine directional changes.

Do not confuse this traditional single-bevel form with a modern double-bevel kiritsuke-gyuto or K-tip gyuto. The silhouette is not enough; bevel construction and handedness must be stated.

4.4 Boning, filleting and fish butchery

4.4.1 Western boning knife

IMAGE PLACEHOLDER

Xinzuo B37-TG Western boning knife

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A Western boning knife is narrow because boning is a constantly changing three-dimensional cut. The edge must enter beside bone, follow joints, turn around curves, lift connective tissue and repeatedly change angle. A broad blade resists these changes once it is engaged in food. **The very small blade**

height is fundamental because it lets the blade change angle easily.

The narrow profile also improves visibility and control while tracing joints, removing connective tissue and separating meat from bone. Stiffer versions give deliberate control around beef and pork; more flexible versions follow rounded surfaces more easily.

Its main limitation is versatility. The narrow blade gives little knuckle clearance, little food-transfer area and less efficiency in broad board work than a chef's knife, gyuto or Chinese cleaver. Recommend it when boning is a recurring task, not as a general-purpose substitute. Boning means working **around** bone, not chopping through it; twisting the point or using it as a lever can still damage the knife.

4.4.2 Honesuki



Figure 4.13: A Xinzuo honesuki example with its triangular poultry-boning profile.

The honesuki is a stiff triangular boning knife used especially for poultry. Its pointed front locates joints and follows bone, while the stronger heel separates connective tissue with controlled pressure.

Its compact length and defined point can also help with selected fish-filleting work, especially the first entry beside bone or under membrane. It does not replace a long flexible fillet knife when the task requires bending around a rib cage. Its strengths are almost the opposite: short reach, rigidity and a highly controllable point.

4.4.3 Fillet knife



Figure 4.14: A Xinzuo Western fillet-knife example, long and narrow enough to follow fish contours.

A Western fillet knife is long, narrow and often flexible. Its thin cross-section lets the blade follow a rib cage, stay close to skin and move around fish contours while leaving less flesh behind. It is a butchery tool that follows the fish's anatomy before presentation cuts, unlike a long rigid sashimi slicer designed mainly for a clean final draw cut.

Flexible models normally use a forgiving balance of toughness and hardness.

A fillet knife made from particularly hard steel needs more caution.

A hard thin blade can flex elastically, but it should never be forced into a deep bend, twisted against hard bone or used to test its flexibility. Excessive lateral stress can cause chipping, cracking or sudden fracture. A break can send the hand and broken edge in an uncontrolled direction, creating a real injury risk.

Use only the flex the model is designed to provide. If the knife feels forced rather than naturally following the contour, change the cutting angle instead of bending harder.

4.4.4 Deba

The deba is a thick, heavy fish-butchery knife. Its strong heel can work through appropriate fish joints and bones with controlled technique, while the front handles finer separation. Traditional deba are single bevelled and handed. Their weight does not make them universal cleavers: large mammal bone, twisting or careless impact can still chip the edge.



Figure 4.15: A Xinzuo deba example, strong at the heel and tapered toward a more precise front.

4.5 Small, table and precision knives

4.5.1 Utility knife



Figure 4.16: A Xinzuo utility-knife example, between a paring knife and a full-size chef's knife.

The utility knife sits between a paring knife and a chef's knife. It has enough reach for fruit, sandwiches, small vegetables, cheese and boneless meat while remaining agile when a full-size blade would crowd the board. Its narrow profile reduces drag but gives less knuckle clearance and food-transfer area than a chef's knife.

4.5.2 Curved paring knife



Figure 4.17: Xinzuo B9H-SG, used here as the curved paring-knife example.

The curved paring knife is short enough to control mainly with the fingers. Its pronounced curve follows rounded fruit and vegetables during peeling, coring, blemish removal and decorative shaping. It is especially effective when the ingredient is held safely and the edge must follow a curved surface.

The same curve gives less straight board contact. Do not treat it as a miniature chef's knife or force it through pits, joints or frozen food.

4.5.3 Straight paring knife

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The straight paring knife places more of its short edge on the board and tracks a direct cut naturally. It works well on garlic, shallots, small fruit, garnish and exact trimming while remaining compact enough for controlled in-hand work. It follows rounded produce less naturally than a curved parer but is more versatile when preparation alternates between hand and board.

4.5.4 Steak knife

A steak knife cuts cooked meat at the table, and the choice between a **serrated** and a **plain** edge changes the eating experience.



Figure 4.18: A Xinzuo table-size steak-knife example.

A serrated steak knife can remain **apparently sharp for a long time**. The serration points keep finding purchase after repeated plate contact, and the teeth can grip and tear fibres after a smooth edge would feel dull. This makes the knife forgiving in everyday service. The trade-off is a rougher cut: serrations partly tear the meat, disturb its texture and can rupture more cells at the surface, allowing more juice to escape. This may be acceptable for inexpensive or casual service but is less suitable for high-quality meat.

For premium steak, a **very sharp plain-edge steak knife** gives a cleaner cut and preserves texture and appearance better. Its weakness is the plate. When every stroke ends against ceramic, the fine apex repeatedly strikes a much harder surface and can lose sharpness quickly.

For valuable meat, the best arrangement is simple: **use a sharp plain-edge steak knife and complete the cut on a wooden plate or small food-safe wooden board rather than directly on ceramic**. If ceramic tableware is unavoidable, the edge will need more frequent maintenance.

4.5.5 Cheese knife

Cheese knives are a family, not one universal shape. Soft sticky cheese benefits from less contact area; a narrow point can portion and serve; hard aged cheese needs stronger support. Openings, hollows and other relief can reduce sticking but cannot eliminate it for every cheese. Choose according to the cheese's texture and size, not the category name alone.



Figure 4.19: A Xinzuo cheese-knife example; different cheese textures still call for different handling.

4.6 Serrated and hard-food specialists

4.6.1 Bread knife



Figure 4.20: A Xinzuo bread-knife example with a long serrated edge.

Bread-knife serrations solve a simple problem: entering a resistant surface and cutting a soft interior **without compressing it first**. Historically, bread-making, storage and common loaf styles made this especially useful, but the principle still applies today.

The teeth grip the surface so the user can make a light drawing or sawing

motion instead of pressing hard downward. Several small points of purchase let the blade advance with less crushing of the crumb or other soft structure.

A pronounced saw profile is **not always necessary** for modern bread. Crust, crumb and freshness vary, and a very sharp plain edge can cut many loaves cleanly. Serrations are most useful when the surface resists first contact but the interior deforms easily. They also work extremely well on brioche, panettone, soft rolls and cake, where downward pressure can deform the product before a plain edge starts the cut.

Use a light guided sawing motion and let the teeth work. Excessive pressure crushes the crumb. Serrations often stay functional for a long time because their recesses receive less direct board contact, but sharpening is less straightforward and may require a tapered sharpener or specialist service.

4.6.2 Frozen-food knife



Figure 4.21: A Xinzuo frozen-food-knife example with a coarse scalloped edge.

A frozen-food knife uses strong coarse serrations to saw through suitable frozen ingredients. It exists because forcing a thin hard chef's knife into a frozen block can chip or break the blade. This is still a hazardous specialist operation: stabilize the food, keep the free hand outside the cutting path and avoid twisting or impact.

4.7 Choosing by work rather than by name

For most kitchens, the first serious knife should be a chef's knife, gyuto, santoku or Chinese cleaver whose length and movement feel natural. A bread knife and

a small utility or paring knife then cover jobs the main blade handles poorly. Long carving, filleting, fish-butchery and impact tools become worthwhile when those tasks occur often enough to justify dedicated geometry.

For sales staff, return to the same questions: **What foods will the customer cut most often? In what quantities? On a board or in the hand? Does the work need broad contact, a long draw, controlled flexibility, a precise point, constant angle changes around bone, or supported mass for impact?** Once the work is clear, the right family of shapes is usually clear too.

Then compare ergonomic fit in the customer's normal grip. The next chapter explains why handle grip, intermediate grip and pinch grip can make the same knife feel different, and why Xinzuo offers different balance strategies rather than one universal setup.

Choose the profile first and ergonomic fit second. Steel, heat treatment, construction, handle material, appearance and price then complete the choice among models that already suit both task and user.

The photographs in this chapter show real Xinzuo versions of the forms discussed. They establish the visible profile, not every property of the knife. Hardness, grind, handedness, heat treatment, balance and bone capability must still be checked on the exact model and understood through the principles in the following chapters.

Chapter 5

Weight, Balance and Ergonomics

Two knives can use equally good steel and careful grinding yet feel completely different in the hand. One feels quick and almost weightless; another feels planted and authoritative. The difference comes from **how much mass the knife carries, where that mass sits and how the handle places the hand in relation to it.**

These are part of the knife's performance. Weight, balance and handle geometry change the movement itself.

5.1 Weight changes the gesture

As soon as we pick up an object, the body prepares for its expected weight. Human motor control anticipates that weight and scales grip and load forces so the object can be lifted and controlled smoothly. A heavier object changes muscular preparation before the movement is fully under way. [4]

The effect is easy to feel with a knife. **A heavier knife encourages a firmer, more committed stroke.** The user prepares to control more mass, while the knife's weight, gravity and inertia contribute to the movement. Once moving, the greater mass also carries more momentum through resistance.

This is why a substantial chef's knife, santoku, deba or Chinese cleaver can make dense food such as squash feel easier to cut than a very light knife. The hand and arm prepare a stronger movement, and the knife's mass helps carry

it through the food. The practical sensation is **authority**: less of the cut feels dependent on forcing a light blade downward.

More mass also takes more effort to lift, accelerate, stop and redirect. A lighter knife feels quicker, needs less effort for repeated changes of direction and is often more comfortable for delicate tip work, rapid preparation and small corrections.

The distinction is simple:

- **More weight** gives a more planted feel and more authority in resistant food, but the hand must control more mass.
- **Less weight** gives more agility and easier redirection, but the user supplies more of the force when food strongly resists the cut.

Geometry still determines how much resistance the food creates. Sharpness, edge angle and thickness behind the edge can change cutting force dramatically, so an efficient light knife can cut more easily than a heavy knife with poor geometry. [1][2] Weight changes how the knife and body attack resistance; geometry changes the resistance itself.

5.2 Where the weight sits changes the knife

Total weight tells only part of the story. Two knives can weigh the same but feel very different because their centres of mass are in different places.

A **blade-forward** knife carries more mass ahead of the hand. The front feels planted and naturally favours downward or push cutting. This suits cooks who like the knife to contribute strongly to the stroke, especially with dense vegetables and repetitive board work.

A **neutral** knife places the centre of mass close to the working grip. It neither pulls strongly forward nor feels anchored at the rear. The hand can guide it without constantly correcting a strong front- or rearward tendency.

More mass toward the **handle** makes the front easier to lift, place and redirect. This can feel quick during fine work and reduce the forward pull at the wrist when the knife is held farther back.

Mass near the tip affects handling more strongly than the same mass close to the hand. The farther mass sits from the grip, the more the knife resists being started, stopped or turned. This is the practical meaning of rotational inertia.

A santoku and bunka show the difference clearly. A typical santoku keeps more steel toward the front, giving the nose a planted feel. A bunka removes more

material around its fine K-tip, trading some front-end authority for visibility and precision. Both are useful because they favour different movements.

5.3 Balance depends on the grip

Every knife has a physical centre of mass, but **how balanced it feels depends on where the hand holds it**. Moving the hand forward or backward changes the leverage between the hand and the mass of the blade and handle.

With a **handle grip**, most of the hand stays behind the bolster. Handle shape, length and rearward mass strongly affect comfort, while a blade-forward design feels more pronounced because the hand supports it from farther back.

With an **intermediate or hybrid grip**, the hand moves toward the bolster. The thumb and index finger may rest on the bolster, at its front or slightly onto the blade. This gives direct blade control while still allowing small shifts backward for force or forward for precision.

With a **pinch grip**, the thumb and index finger hold the blade faces just ahead of the bolster. The controlling fingers sit much closer to the blade. A knife that feels strongly blade-heavy from the rear handle can feel almost neutral when held in a forward pinch grip.

This also explains why some handles that look unusually long in photographs feel natural in the hand. The rear section can act as a counterweight, moving the balance back toward the working grip even when the cook holds mainly the front of the handle.

People develop different habits, so there is no single correct balance for every user. **Xinzuo deliberately produces knives with different balance strategies for different grip positions**. Some designs are intended to feel natural in a forward pinch grip, some suit an intermediate grip around the bolster, and others are balanced for users who hold mainly the handle. The purpose of this variety is not to create a ranking from better to worse. It lets different users find a knife whose mass distribution matches the way they already control the blade.

Do not ask only, **“Is this knife well balanced?”** Ask, **“How does this knife balance when this customer holds it the way they actually cook?”**

5.4 Ergonomics is part of performance

Once the blade shape suits the work, ergonomics becomes the next selection criterion. Handle shape, circumference, length, contour, bolster transition, total weight and balance determine how naturally the knife fits a particular hand and movement.

There is no substitute for handling the knife. Palm width, finger length, grip position and habit vary from person to person. Two knives with similar photographs and specifications can feel very different because a few millimetres of handle contour or a small change in balance alters where the fingers settle and how the wrist supports the blade.

This makes a physical store especially valuable. A catalogue can identify technically suitable knives; **the customer's hand can identify which one has the best ergonomics**. Choosing a knife is similar to choosing a pair of shoes: specifications narrow the field, but fit decides which suitable option feels right.

Xinzuo's broad portfolio is a major advantage here. Its range of blade forms, handle geometries, weights and balances lets staff identify several knives that meet the customer's functional needs and then find the one that best fits the person's hand, grip and cutting style.

5.5 How staff should guide the choice

For an in-store recommendation:

1. **Identify the work.** Ask what the customer cuts most often, in what quantities and with which movements.
2. **Choose the blade family and shape.** Solve the functional problem first: chef's knife, santoku, bunka, nakiri, carving knife, boning knife, cleaver or another specialist form.
3. **Identify the natural grip.** Let the customer hold the knife as they normally cook. Determine whether they mainly use a handle grip, an intermediate grip or a pinch grip.
4. **Compare handle feel and balance in that grip.** Test handle shapes and sizes, then compare blade-forward, neutral and more handle-weighted designs while the customer keeps their normal grip.
5. **Offer a shortlist.** Put several knives that meet both the functional and ergonomic needs into the customer's hand and let them compare directly.
6. **Use steel, construction, materials, appearance and price to complete the choice.** These factors distinguish between knives that already suit the customer's use.

Remember the method as **function, fit, selection**. First find the shape that does the job. Then find the ergonomics that suit the person. Finally choose the specific product among the models that satisfy both.

Chapter 6

Single and Double Bevels

Cut a tall potato with two knives and you may notice something unexpected. One blade falls almost straight; another pulls sideways even when your hand points forward. The bevel—the sloping surface leading to the cutting edge—may explain the difference.

Three parts are often confused.

The **primary bevel** is the broad ground surface that reduces blade thickness toward the edge. Near the edge, many kitchen knives have a much smaller terminal facet. This book calls that narrow sharpening surface the **microbevel**. The microbevels—or the microbevel and opposite face in some asymmetric constructions—meet at the **apex**, the microscopic line that cuts.

This matters because routine resharpenering of a double-bevel knife works on the microbevel. Laying the entire broad primary bevel on the stone changes the geometry behind the edge and belongs to thinning, geometry correction or deliberate reprofiling. The microbevel is not always easy to see, so the practical reference is the small terminal facet immediately behind the apex that the stone must reach.

Bevel geometry changes four things a cook can feel:

- the direction the blade tends to travel;
- how the cut piece separates from the food;
- how naturally the knife works in the right or left hand;
- how the knife must be sharpened.

The two broad families are double bevel and single bevel. Neither is automati-

cally sharper or more advanced. They solve different problems.

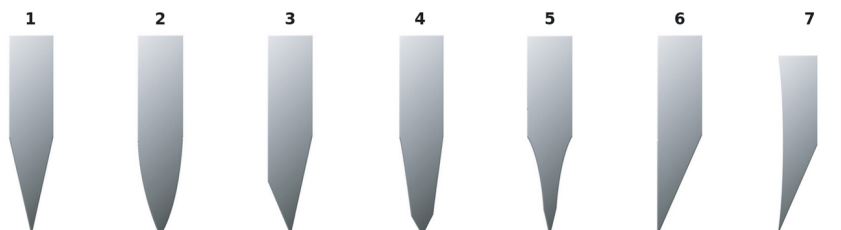


Figure 6.1: Seven common bevel families, shown without fixed-language labels: 1. symmetrical V; 2. convex; 3. asymmetric V; 4. compound double V; 5. concave; 6. single-sided; 7. single-sided with urasuki. The legend is ordinary text so it can be translated with the book.

6.1 Double bevel: the familiar all-purpose edge

A double-bevel knife is sharpened from both sides. Seen from the front, the sides converge toward a narrow V-shaped edge. On many knives the broad blade grind and tiny microbevel are visibly separate; on others the transition is subtle.

A reasonably symmetrical double bevel tends to track straight because food is displaced on both sides. This helps make it suitable for general cooking and either hand.

Most Western chef’s knives, carving and slicing knives, utility and petty knives, gyuto, santoku, bunka, nakiri and many Chinese cleavers and slicers are double bevel.

6.1.1 How it feels during a cut

Make a slow push cut through a potato or large carrot. If the edge is symmetrical and the blade is not unusually thick, it should follow your hand without much correction. The food may stick to the blade, but neither side strongly dictates the knife’s direction.

This predictability is useful in a first serious knife. It lets the user learn grip, board control and movement without also managing strong steering.

6.2 Not every double bevel is symmetrical

The edge may be close to 50/50 or biased toward one face. Labels such as 60/40 or 70/30 are useful descriptions but are often approximate rather than exact measurements.

An asymmetric double bevel moves the cutting edge closer to one side. This can change food separation and make some fine cuts feel more precise, but it can also introduce slight steering.

Most mildly asymmetric knives remain usable in either hand. When sharpening, preserve the existing geometry instead of automatically giving both sides the same number of strokes.

6.3 Single bevel: one side does most of the shaping

A traditional single-bevel knife has a broad main bevel on one face. The reverse is much flatter and often contains a shallow hollow called the **urasuki**.

The edge sits close to one side of the blade rather than near the centre. As the main bevel enters food, it pushes the cut portion away while the flatter face can guide the piece that remains.

Common examples include:

- yanagiba for finished slices of raw fish;
- usuba for precise vegetable work;
- deba for fish butchery;
- selected kiritsuke, mukimono and other traditional specialist forms.

A traditional profile name does not prove its grind. Modern manufacturers also make double-bevel versions of some traditionally single-bevel forms. Check the exact Xinzuo model before describing handedness or sharpening.

6.3.1 The main bevel may change angle from heel to tip

A single-bevel blade is three-dimensional. Blade height decreases as the outline narrows toward the tip, and the spine may also become thinner through **distal taper**, the gradual reduction in blade thickness toward the tip. These changes alter the relationship between blade thickness, bevel width and edge position.

The local angle of the broad single bevel **can change along the blade**. Do not assume the heel, middle and tip form one perfectly uniform plane with

one theoretical angle, especially on knives with strong distal taper or a rapidly narrowing point.

During primary-bevel sharpening, use actual stone contact as the guide. The knife position may need to change gradually toward the tip so each local section of the bevel stays supported. Forcing one wrist position from heel to tip can miss part of the bevel, overwork another or alter the intended geometry.

Sharpen slowly and inspect often. Watch where the stone removes steel, use marker or reflected light when useful, and preserve the geometry already present instead of imposing an abstract constant angle.

6.4 The urasuki is not an unfinished back

The shallow hollow on the reverse reduces contact with food and the sharpening stone. Narrow flat areas remain around it as reference surfaces.

Do not sharpen this side like the second face of a normal V-edge. It is usually placed almost flat on the stone and worked lightly. Raising it to a normal sharpening angle can remove the reference surface, alter the edge and gradually destroy the hollow's function.

For a beginner, remember:

A single bevel is a complete three-dimensional system, not an ordinary edge with one side forgotten.

6.5 Handedness and working direction

The broad bevel determines how food separates and where the flatter face sits. Changing hands does not reverse these forces.

A right-handed yanagiba is arranged so that, in its intended position, the bevel moves the cut slice away while the flatter face guides the remaining food. A left-handed version mirrors the geometry.

The matching version usually feels more natural because the knife, body, supporting hand and cutting direction work together. This matters during frequent professional use.

A knife made for the opposite hand can still be used, but the food and cutting direction must be arranged around its geometry. For example, a left-handed cook using a right-handed knife can turn the ingredient and choose a path that

carries the edge away from the supporting right hand. The stroke can travel sideways into clear space or forward and away from the body.

Changing the setup works better than overpowering the blade. Keep the supporting hand behind the cutting line, leave the path clear and keep the wrist relaxed. If the knife drifts, stop, rotate the ingredient or adjust your stance, then restart. Forcing a buried edge sideways stresses the apex and reduces control.

Learn this with a short, stable piece of cucumber or a small boneless portion. A tall potato, hard squash or rolling ingredient magnifies steering and encourages excess force. Slow, shallow cuts reveal the knife's tendency before it is deep in the food.

This approach works when the ingredient can be rotated freely. A knife matched to the user's hand remains more convenient during long professional sessions or repeated identical cuts because the workstation does not need to be reorganized for each piece.

6.6 How steering develops

Food pushes back as the blade enters. With a symmetrical V-edge, pressure is more evenly distributed. With a single bevel, one face acts as a wedge while the other remains much flatter. This imbalance pushes the blade sideways.

Steering becomes more noticeable when:

- the ingredient is tall, hard or dense;
- the blade is thick;
- the bevel is broad;
- the cut is long;
- the user applies excessive downward force;
- the wrist tries to correct after the blade is already deep in the food.

The best correction begins before the cut: choose the direction, stabilize the food and set the starting angle. Large corrections are risky once the blade is deeply buried because they can twist the edge.

6.7 Clean surfaces and food separation

A single bevel can leave an exceptionally clean surface because the bevel directs the cut piece away while the flatter face guides the remaining food. This

matters in sashimi, where surface quality affects appearance and texture, and in katsuramuki, where an usuba turns a vegetable into a continuous thin sheet.

A double-bevel carving knife can also cut extremely cleanly. Long, narrow carving knives are easier for many cooks, work in either hand and handle boneless meat and fish very well.

The useful distinction is not “single bevel is sharp and double bevel is ordinary.” Both can reach a very fine apex. They differ in steering, separation, surface finish and specialization.

6.8 Strength does not come from bevel count alone

A deba is single bevel and thick because it is built for fish butchery. An usuba is also single bevel but has a much finer working edge. A heavy double-bevel chopper is robust, while a thin double-bevel gyuto can chip if twisted.

To judge how demanding a knife will be, consider:

- blade thickness;
- thickness behind the edge;
- edge angle;
- steel and hardness;
- tip shape;
- intended task.

Bevel type is only one factor.

6.9 Choosing between the two

Choose a double bevel when you want:

- one knife for varied work;
- easy use in either hand;
- predictable straight tracking;
- a familiar sharpening process.

Consider a single bevel when you want:

- a specialist result such as sashimi slicing or traditional vegetable work;
- controlled food separation;
- the surface finish associated with the intended technique;
- a knife you are willing to learn and maintain as a specialist tool.

Within Xinzuo's range, compare the exact model rather than relying on the profile name. Verify the grind, handedness, steel, thickness and intended food.

6.10 What the geometry means for sharpening

Sharpen the geometry that is actually on the knife instead of applying one rule to every edge. On an ordinary double bevel, routine sharpening works the narrow microbevel on both sides. Preserve intentional asymmetry instead of automatically converting the edge to 50/50.

A traditional single bevel is different. The broad main bevel forms part of the sharpening geometry, while the reverse is usually worked almost flat to preserve the urasuki and its reference flats. Contact on the main bevel may also change toward the tip as blade height and thickness change.

The beginner sequence appears later in *Your First Successful Hand Sharpening*. For now, remember: **identify the geometry first, then sharpen the surfaces that define it.**

6.11 What to remember

- The broad primary bevel and narrow microbevel are different surfaces.
- Double bevel means sharpening from both sides, not necessarily perfect symmetry.
- Single bevel is a complete geometry, often including an urasuki.
- On a single bevel, the local main-bevel angle can change as blade height and thickness change toward the tip.
- A single-bevel knife is optimized for one hand but can often be used with the other when the food and cutting direction are arranged around the geometry.
- Manage steering through setup and direction, not forced sideways correction.
- Neither family is automatically sharper, stronger or more prestigious.

Practice principle: preserve the geometry that is actually on the knife.

Chapter 7

How to Cut - Essential Knife Techniques

The right knife helps, but movement matters even more. Good technique makes pieces more even, protects the edge, reduces fatigue and keeps the supporting hand safer. It also explains why two equally sharp knives can feel very different on the same board.

Learn one movement at a time, slowly enough to see what the blade is doing. Speed should come from familiarity, never be the first goal.

This is knife gongfu in practice: a small set of controlled movements repeated until they become natural without losing awareness.

7.1 Before the first cut

Keep the board from sliding with a damp cloth or non-slip mat underneath. It should be large enough for the ingredient, the knife and a clear area for cut pieces.

Stand close enough that you do not need to reach. Relax your shoulders and keep the knife wrist in a natural line with the forearm. If the wrist must bend sharply, move the ingredient or your body instead.

Hold the food with the supporting hand in a **claw**: fingertips curved inward, thumb behind them and the side of the blade guided by the first joints instead

of exposed fingertips. The claw is a moving guide, not a rigid pose; it retreats slightly after each cut.

The three main knife-hand positions—**handle grip, intermediate or hybrid grip, and pinch grip**—are explained in Weight and Balance. Choose the grip that feels stable on the knife, keep it relaxed and remember that moving the hand forward or backward changes the balance you feel.

The four-word safety check: stable board, stable food, clear path, hidden fingertips.

7.2 Read the path before naming the technique

Kitchen movements are not four sealed categories. A real cut can combine vertical travel, forward or backward travel, and a changing contact point. Read the path of the edge:

What the edge does	What that movement solves	Profiles that often help
Meets the board across a broad, flat section	Regular vegetable pieces and direct board work	Nakiri, flatter santoku, bunka or gyuto
Advances while descending	Reduces wedging and uses a short slicing action	Gyuto, santoku, bunka and many general-purpose knives
Travels along its length through the food	Crosses wide or delicate food with fewer interruptions	Carving and sashimi slicers
Keeps one area near the board while another rises	Repeats mincing work through a controlled arc	Curved chef's knives or suitable round-front profiles

The names below describe dominant movements, not rigid rules. Choose the path that lets the edge pass through the ingredient with control and without forcing the blade to imitate a geometry it does not have.

7.3 Board contact: direct descent

7.3.1 What the movement looks like

The blade rises and falls in a mostly vertical path. It may move slightly forward as it descends, but it does not travel through a large arc. A broad, relatively flat section of edge meets the board.

This suits nakiri, santoku, bunka, many gyuto and Chinese cleavers. A strongly curved Western chef's knife can also chop, but part of its belly may stay above the board in a perfectly vertical stroke.

7.3.2 Try it with a cucumber

1. Cut a thin strip from one side so the cucumber cannot roll.
2. Place the flat side on the board.
3. Form the claw with the supporting hand.
4. Raise the knife only enough to clear the cucumber.
5. Lower it in a controlled line and let the edge finish the cut.
6. Move the claw back by the desired thickness and repeat.

The knife should make light, regular contact with the board, not heavy impacts. If pieces remain joined by thin skins, the edge may be too curved for the movement. Add a slight forward motion or use a flatter section of the blade.

7.3.3 Why it works

A fairly straight edge can contact the board along much of its length at once. This makes straight chopping efficient for vegetables and regular pieces. A tall blade also gives the guiding hand a large, predictable surface.

7.3.4 Common mistake

Do not confuse chopping with striking. Lifting the knife high and driving it down adds noise and risk without improving an ordinary vegetable cut.

7.4 Board contact: advancing stroke

7.4.1 What the movement looks like

The knife moves forward and downward together. The edge enters near the front or middle, travels through the ingredient and finishes with the heel or

middle touching the board. The knife then lifts and returns without dragging the edge sideways.

This movement feels natural with a moderately flat gyuto and also works well with santoku, bunka and nakiri.

7.4.2 Try it with a carrot

Cut a small flat on the carrot so it rests securely. Start with the front half of the edge above the food. Move the knife forward as it descends. Think of drawing a short diagonal line through the carrot rather than dropping the blade straight down.

If the carrot cracks before the edge passes through, use less force and make sure the knife moves forward as well as down. A thin edge cuts by concentrating force and sliding; it should not need to split the vegetable like a wedge.

7.4.3 Why it works

An edge cuts more efficiently when it slides across food. The forward component gives each point of the edge a short slicing journey instead of asking it to press straight through. A flatter blade completes the cut without a large rocking motion.

7.4.4 A memory aid

Think **forward, down, release**. Forward starts the slice, down completes it, and release lifts the edge instead of scraping it across the board.

7.5 Long work: draw through the food

7.5.1 What the movement looks like

The heel or rear of the knife starts near the ingredient and the blade travels toward the user as it cuts. A long carving knife can use most of its edge in one smooth draw.

This is the natural movement of carving knives. A long double-bevel carving knife is a versatile slicer for boneless meat and fish; the yanagiba is a specialist single-bevel slicer for raw fish. A chef's knife can use the same movement for boneless meat or a ripe tomato.

7.5.2 Try it with a boneless cooked chicken breast

Place the breast across the board. Set the heel just beyond the far side. Use light pressure and draw the blade toward you in one long movement, keeping the path clear of the supporting hand and body. If the slice is incomplete, reset and make another long pass. Avoid short rapid sawing unless the food or a serrated edge requires it.

7.5.3 Why it works

A long draw spreads the work across more of the edge. It reduces downward compression and leaves a smoother cut surface. This matters with delicate fish because surface quality affects both appearance and texture.

Traditional sashimi knives are long so the cook can finish a slice with few interruptions. The single-bevel yanagiba adds controlled food separation to that long draw. The surface left by the cut becomes part of the finished dish.

7.6 Pivoted work: keep a point of contact

Two related movements should be separated: the **wide rolling rock** of a curved chef's knife and the **compact pivot rock** of a round-tip nakiri.

7.6.1 Wide rolling rock

With a Western chef's knife, the front of the edge stays on or near the board while the heel rises and falls. The contact point travels along the curved belly like part of a wheel. Some gyuto also rock well; the actual profile matters more than the label.

This movement depends on curvature. Do not force a very flat blade with no suitable front transition to imitate it.

7.6.2 Why the round-tip nakiri suits a compact rock

A **round-tip nakiri** solves the same problem with different geometry. Its rounded front, rather than a long belly, can stay in controlled contact with the board while the heel rises and falls through a short arc. The extreme nose is normally unsharpened or transitions safely away from a piercing point.

For compact rocking, this geometry is close to ideal. The rounded nose creates a broad, predictable pivot instead of loading a fragile point. The tall blade gives knuckle clearance and a clear guide for the non-cutting hand. Behind

the pivot, the relatively straight edge lands decisively on the board, cutting herbs and small ingredients cleanly instead of leaving them connected under an overly curved belly.

Use a **short, controlled pivoting movement**, not the exaggerated sweep of a classic chef's knife. Keep the front planted, raise the heel only as much as needed and advance through the pile in small steps. This works especially well for herbs, fine vegetable mincing and repeated compact board work.

The rounded front is functional, not simply a tip with less piercing ability. It creates a stable board-contact point that tolerates repeated pivoting better than a delicate K-tip.

7.6.3 Try it with parsley

Gather the leaves into a compact pile. With a curved chef's knife, place the forward working section on the board and roll across the belly. With a round-tip nakiri, use the rounded front as the pivot and raise and lower the heel through a short arc. Keep the guiding hand safely on the spine or upper blade area, never near the edge. Re-gather the herbs with the broad blade face, not the edge.

7.6.4 Common mistake

Do not rock on the fragile K-tip of a bunka or kiritsuke-style knife. A K-tip is a precision point, not a bearing surface for repeated board contact or lateral stress. Do not force a square-front or very flat nakiri into a wide rolling arc either.

7.7 Tip work

Use the tip for small incisions, trimming and confined spaces. Keep the wrist relaxed and move the food when possible instead of twisting the point inside it.

To core a tomato, stabilize it on the board, insert only the needed length of tip beside the stem and guide a shallow circle. To score skin or pastry, use light pressure and a drawing motion.

The pointed tip of a gyuto or chef's knife is versatile. A bunka's K-tip gives excellent visibility but is usually thinner and more vulnerable. A round-tip nakiri deliberately gives up piercing work for a robust, predictable front that supports compact pivot cutting.

Rule: a tip cuts by moving along its edge. It is not a screwdriver, lever or drill.

7.8 Slicing across the grain

Muscle fibres run in visible bundles. Cutting parallel to them leaves long fibres that are harder to chew; cutting across them shortens the fibres and usually makes the same meat feel more tender.

Before cutting, identify the grain and rotate the meat so the knife crosses it. Use a draw slice with a carving knife or a long chef's knife. The improvement comes from direction, not extra pressure.

This is a good example of knife skill beginning before the blade touches the food.

7.9 Fine slicing and julienne

Julienne means cutting food into thin, even strips similar to matchsticks. Uniform pieces cook at a similar rate, which is the practical reason to learn the technique; appearance is only part of the benefit.

For carrot julienne:

1. Make one flat side and place it on the board.
2. Cut the carrot into even slabs.
3. Stack only as many slabs as remain completely stable.
4. Cut the stack into even sticks.

Use push cuts or controlled straight chops. Let the claw knuckles guide the blade side. Do not race the supporting hand. Move the knife only after the claw is set.

A Chinese cleaver can work especially well here. Its tall blade provides a large guide surface, while a thin Chinese vegetable cleaver, often called a **cai dao**, can make very fine slices. Its rectangular shape does not make it a bone cleaver.

7.10 Dicing

Dicing means cutting food into roughly cube-shaped pieces. It is a sequence of controlled cuts in different directions. Create a stable shape, make parallel slices, rotate or restack the food, then cut across them.

With an onion, leave the root end intact during the first cuts so the layers stay connected. Keep the guiding hand above and behind the edge. The exact classical method matters less than three rules: keep the onion stable, hide the fingertips and control the final cross-cuts.

If the dice vary greatly, slow down and improve the first slices. Irregular dice usually begin before the final cut.

7.11 Mincing without crushing

Mincing means reducing food to very small pieces. Start with slices, then strips, then cross-cuts. Gather the pieces and make extra passes only after they are already small.

For garlic or herbs, use straight chopping with a compact blade, a gentle rolling action with a curved chef's knife, or the short pivot rock of a round-tip nakiri. Re-gather food with the spine or broad face. Repeatedly scraping the edge sideways against the board weakens the apex.

Do not chase an arbitrary number of passes. Stop when the texture suits the recipe. Bruised herbs and garlic paste can be desirable, but they are different results from a clean mince.

7.12 Crushing and transferring food

The broad face of a chef's knife or Chinese cleaver can crush a peeled garlic clove. Lay the blade flat with the edge facing away from the body, place the heel of the free hand on the broad face and press once with control. Never strike the blade while fingers are near the edge.

To transfer chopped food, slide the broad face underneath or use a bench scraper. If you use the knife, lift with the blade face while keeping the edge clear of hands. Do not drag the cutting edge sideways across the board.

7.13 Peeling and in-hand work

A short paring knife can cut food held in the supporting hand, but the movement must be small and deliberate. Hold the ingredient securely and use a shallow path that finishes beside the thumb, not into it. The thumb guides and brakes the food; it must never sit in the path of an uncontrolled stroke.

Beginners should first practise peeling on the board or use a peeler. Large chef's knives, long carving knives and heavy Chinese knives are not suitable for casual in-hand cutting.

7.14 Working around bones and joints

Boning is usually tracing, not chopping. Use the point to locate the joint and the edge to separate connective tissue. A Western boning knife follows contours; a honesuki uses a stiff triangular profile for poultry; a deba combines a strong heel with specialist fish geometry.

Do not twist a fine edge against bone. If the joint does not separate, stop and find the gap instead of adding force. A knife designed to work around bone is not automatically designed to cut through it.

7.15 Filleting fish

Stabilize the fish and follow a clear sequence: open it, locate the backbone, let the blade follow the bones and use long controlled strokes to separate the fillet. A flexible fillet knife follows contours; a traditional deba and yanagiba divide the work differently, with the deba used for butchery and the yanagiba for final slicing.

The blade should tell you where the skeleton is. Heavy pressure hides that feedback and wastes flesh. Learn slowly enough to feel contact without forcing the edge into hard bone.

Traditional single-bevel geometry also affects steering and food separation. Single and Double Bevels explains how to arrange food and cutting direction when using one with the opposite hand.

7.16 Specialist techniques that need demonstration

Some techniques cannot be learned safely from a short written description. **Katsuramuki**, the technique of peeling a vegetable into a long continuous thin sheet with an usuba, fish-head work with a deba and rapid professional chopping combine blade geometry, body position and practiced feedback. This guide explains the principles; a qualified demonstration is the safest way to learn the movement itself.

Cooking is learned through words, observation and repetition. The page provides the map; demonstration provides body position, timing and feedback that are difficult to communicate safely in text.

7.17 A seven-day practice sequence

Use inexpensive, stable ingredients and stop before fatigue reduces control.

Day	Practice	What to observe
1	Cucumber, straight chopping	Stable board and complete edge contact
2	Carrot, push cutting	Forward movement before extra force
3	Cooked boneless food, draw slicing	Long stroke and smooth cut surface
4	Parsley, round-tip nakiri pivot or chef-knife rock	Stable pivot and natural use of the actual profile
5	Carrot or celery, julienne	Uniform first slices and claw position
6	Onion, dice and mince	Stable sequence rather than speed
7	Repeat the least comfortable movement	Relaxed grip and consistent pieces

7.18 What to remember

- Choose and understand the grip in the weight-and-balance chapter; do not relearn the same grip taxonomy here.
- Straight chopping uses a flat section of edge.
- Push cutting moves forward and down.
- Draw slicing uses blade length and minimizes compression.
- A curved chef's knife suits a wide rolling rock; a round-tip nakiri suits a short, stable pivot rock.
- Uniform preparation begins by stabilizing and orienting the food.
- Specialist knives reward the movement they were designed for.
- Speed is the result of control, not a substitute for it.

Practice principle: control first, consistency second, speed last.

Chapter 8

The Five Dimensions of Knife Steel

Begin with the work the edge must perform. A long slicing session, repeated contact with acidic food and occasional work near joints place different demands on steel. HRC, edge retention and alloy price describe only parts of that problem.

One steel may hold an edge for a long time but require demanding abrasives. Another may sharpen beautifully but need more frequent maintenance. A highly corrosion-resistant steel may suit a busy kitchen, while a reactive carbon steel may suit someone who values sharpening response and is willing to clean and dry the blade immediately.

The useful question is:

Which balance of qualities best suits this knife, this task and this user?

Five dimensions provide a practical map: **hardness, sharpening response, corrosion resistance, toughness and microstructural fineness**. *Microstructure* means the microscopic internal structure of the steel: its tiny crystals, phases and hard particles. The later metallurgy chapters will explain these parts in more detail.

8.1 1. Hardness: resistance to permanent deformation

Hardness is resistance to indentation and permanent deformation. Kitchen-knife hardness is commonly reported on the Rockwell C scale, abbreviated **HRC**.

HRC does not measure sharpness, toughness or overall cutting performance. A knife is sharp because its edge is geometrically fine and clean. Hardness helps determine how that geometry changes during use.

A harder apex generally resists rolling and plastic deformation better. Hardness also contributes to abrasive wear resistance. With broadly similar steel composition and microstructure, the harder sample usually resists wear more strongly.

Steel can also contain **carbides: microscopic hard crystals formed when carbon combines with iron or certain alloying elements**. Carbide type, amount, size and distribution can change wear resistance greatly. Vanadium- or niobium-rich carbides, for example, are much harder than the surrounding steel matrix.

The **matrix** is the continuous part of the steel that surrounds these particles. Its own hardness matters too. Two steels at the same HRC can therefore behave differently because their matrices and carbide populations are different.

The practical rule is simple: **do not rank unrelated steels by HRC alone**. Read hardness together with heat treatment, microstructure, edge angle, thickness behind the edge and intended use.

Higher hardness can also reduce tolerance for misuse. A very hard, thin edge may remain extremely stable in correct cutting yet tolerate twisting, levering or impact poorly. Hardness works best when the steel's toughness and blade geometry suit the intended task.

8.2 2. Sharpening response: more than “soft equals easy”

Sharpening response describes how readily and predictably an abrasive can restore the apex. Hardness is only part of the picture.

A harder matrix resists abrasion more strongly, but carbides can matter even more. Large volumes of very hard carbides slow material removal and may

need efficient ceramic or diamond abrasives. Burr behaviour matters too. One steel may form a small, crisp burr that releases cleanly; another may form a persistent wire edge that takes more work to remove.

The shortcut “**softer steel is easier to sharpen**” is **unreliable**. A moderately soft stainless steel may be forgiving in use without feeling crisp on a water stone. A well heat-treated high-carbon steel can run at much higher HRC yet abrade quickly, give clear feedback and deburr cleanly.

DIN 1.4116 shows the difference well. Kitchen knives often use it at moderate hardness, which can make them forgiving in everyday use. That does not make the steel inherently easy to sharpen. Depending on heat treatment, geometry and abrasive, it can develop a tenacious burr and feel less clean on the stone than a fine, simple carbon steel. **Blue Steel No. 2 (Aogami #2) can respond very readily to suitable water stones even though it is commonly used at much higher hardness.**

Keep two ideas separate:

- **forgiving use:** how readily the edge tolerates deformation or ordinary mistakes;
- **sharpening response:** how efficiently the steel abrades, reaches the apex and deburrs.

They affect each other, but they are not the same property.

8.3 3. Corrosion resistance: what “stainless” means

Corrosion resistance is a blade’s ability to resist moisture, oxygen, salt, acids and other substances encountered in use.

The widely used compositional threshold for stainless steel is **at least 10.5% chromium by mass**. This classifies the steel; it does not mean every steel above the threshold resists corrosion equally well.

Chromium works mainly through **passivation**. This means that enough chromium at the surface forms an extremely thin protective oxide layer that slows further corrosion. If scratched, the film can form again when suitable conditions return.

The key word is *available*. Chromium dissolved in the matrix can support passivation; chromium tied up in carbides is less available for that job. Carbon

content, heat treatment, molybdenum, nitrogen, surface finish and the working environment all affect the result.

Stainless steel is corrosion-resistant, not stain-proof. Salt, chlorides, acidic residue and prolonged wet storage can still stain or pit a stainless blade. Reactive carbon steels need faster care and may develop a dark patina, which is different from active orange rust.

8.4 4. Toughness: resistance to fracture

Toughness is the ability to absorb stress and energy without fracturing. At the edge, low toughness may appear as chipping. At a larger scale it can lead to cracking or breakage under impact, twisting or strong lateral loads.

Toughness is narrower than the everyday idea of *robustness*. A finished knife also gains or loses robustness through thickness, edge angle, blade shape and the amount of material supporting the edge.

The same steel can feel delicate in an extremely thin precision slicer and much more tolerant in a thicker working geometry. Even a very tough steel cannot make an ultrathin edge immune to lateral abuse.

The task determines the useful compromise. A sashimi slicer benefits from refinement and low cutting resistance; a bone chopper needs much more support behind the edge.

8.5 5. Microstructural fineness: what supports the apex

Steel looks uniform to the eye, but under a microscope it is made from many tiny crystal **grains** and can contain different phases and carbide crystals. Their size, amount and distribution affect wear resistance, toughness and how finely the edge can be refined.

A finer, more uniform microstructure generally supports a very fine, continuous apex more easily. This does not mean the edge radius must match the diameter of the largest carbide. The mechanism is more complex. The useful point is that very large hard particles can make an extremely smooth, continuous edge harder to create and stabilize.

Large or abundant hard carbides can still be useful. They can increase wear resistance and preserve an aggressive, toothy slicing character for a long time.

The trade-off is usually slower sharpening, while high carbide size or volume can reduce toughness and the potential for an ultra-refined apex.

The term **grain size** is sometimes used loosely for all of this, but grain size and carbide size are different. This guide uses *microstructural fineness* as the broader teaching term. The later metallurgy chapters explain grains, matrix, phases and the main carbide families in more detail.

8.6 The five dimensions interact

The five dimensions are not independent sliders that can all be maximized.

- Higher hardness can improve resistance to rolling and wear while reducing tolerance for some forms of misuse.
- More hard carbides can improve abrasive wear resistance while slowing sharpening and often reducing toughness.
- Strong corrosion resistance depends on alloy balance and chromium available for passivation, not simply on nominal chromium content.
- Very thin geometry can transform cutting performance while demanding more from the steel and the user.
- A steel can sharpen beautifully and still hold an edge well, or sit at moderate hardness and still resist sharpening.

These trade-offs are why different steels and heat treatments exist.

8.7 A worked example: reading the Xinzuo range

The Xinzuo Blade Steels chapter applies this framework to a real product range. ZDP-189 emphasizes very high hardness and wear resistance; SRS13, M390 and 14Cr14MoVNB pursue different premium balances; 10Cr15CoMoV occupies a broad all-round position; Blue Steel No. 2 exchanges stainless convenience for responsive carbon-steel behaviour; and DIN 1.4116 sits at the more forgiving, moderate-hardness end of the range.

These are tendencies, not complete predictions of a finished knife. Heat treatment, grinding, edge geometry and sharpening still determine how a model behaves. Xinzuo's documented use of vacuum heat treatment and deep-freeze processing in selected series shows how manufacturing turns an alloy's potential into a working blade; the dedicated heat-treatment chapter explains those processes.

A home cook may value corrosion resistance and tolerance of ordinary use. A professional making precise fish or vegetable cuts may accept more demanding care for greater hardness, fine geometry and sharpening response. A butcher working around joints may put toughness and blade support ahead of extreme refinement.

None of these priorities is inherently more advanced. They serve different work.

The central lesson is more useful than “higher is better”:

A steel grade describes potential. Heat treatment develops the microstructure, geometry determines how the edge meets food, and the user’s technique determines which trade-offs matter in practice.

Chapter 9

Alloying Elements

Steel is often presented as a list of ingredients. A product page may mention carbon, chromium, molybdenum, vanadium, tungsten or cobalt and imply that a longer list means a better knife.

Steel is closer to a recipe. The result depends on how much of each element is present, how the elements interact, how the steel is made and how it is heat-treated. Two alloys with similar ingredients can behave very differently.

Keep three principles in mind: no alloying element works alone; more is not automatically better; and composition creates possibilities that heat treatment and blade geometry must turn into useful performance.

Xinzuo's range makes these interactions concrete. It includes chromium-rich stainless steels, cobalt-bearing 10Cr15CoMoV, niobium-bearing 14Cr14MoVNb, nitrogen-alloyed 14C28N, tungsten-bearing Blue Steel No. 2 and powder-metallurgy steels such as SRS13, M390 and ZDP-189. Each is a different metallurgical recipe built around a different balance.

9.1 Follow each element through the steel

An alloy percentage is only the starting point. To understand an element, follow four questions: where does it remain after heat treatment, what structure does it help create, which property can that structure support, and what new constraint does it introduce?

Element	Main contribution to look for	Question that prevents oversimplification
Carbon - C	Hardness and carbide formation	How much remains in the iron-rich matrix and how much forms carbides?
Chromium - Cr	Passivation and, in some alloys, carbides	How much remains available for corrosion resistance instead of being tied up in carbides?
Molybdenum - Mo	Hardenability and resistance to localized corrosion	How does the amount and heat treatment change those effects?
Vanadium - V	Grain control and very hard carbides	Is the useful role mainly grain refinement, wear resistance, or both?
Tungsten - W	Wear-resistant carbides and high-temperature stability	Does the added wear resistance justify slower sharpening for this knife?
Cobalt - Co	Support for demanding heat-treatment cycles	Which process window does it help, rather than which property is being inferred from its name?
Nickel - Ni	Toughness in suitable alloys and bright contrast in layered steel	Is it part of the working steel or mainly a contrasting layer?
Nitrogen - N	Hardness and corrosion resistance alongside carbon	How does it share those roles with carbon and chromium?

The sections below answer these questions and connect composition to microstructure, processing and use.

9.2 Carbon: what makes hard knife steel possible

Carbon is the central alloying element in most knife steels. Without enough carbon, iron cannot reach the hardness needed for a durable cutting edge.

A mass percentage can make the amount sound tiny. **0.5% carbon by mass means only 0.5 grams of carbon in 100 grams of steel.** But carbon is much lighter than iron, so this is roughly **one carbon atom for every 43 iron atoms.** A small mass fraction can strongly change the steel.

Steel is made from microscopic iron crystals. Carbon atoms are much smaller than iron atoms, so they can fit into tiny spaces between iron atoms inside those crystals. They are not simply foreign particles mixed into the metal.

The amount of carbon that can remain in these spaces changes with temperature and with the internal crystal structure of the iron. During hardening, heating creates **austenite**, a high-temperature form of the iron-rich steel that can hold much more carbon. Rapid cooling can then produce **martensite**, the very hard structure used in a hardened knife. Martensite is created so quickly that more carbon remains trapped in the iron structure than it would under equilibrium conditions.

Austenite and martensite are not carbides. In both, iron forms the main crystal structure and carbon is dissolved within it. A carbide is different: it is a separate crystalline compound in which carbon is chemically combined with iron or another carbide-forming element. **Cementite (Fe_3C), for example, is a true iron carbide.** Chromium, vanadium, molybdenum, tungsten and niobium can also take part in carbide formation in knife steels.

This gives carbon two very different roles in steel. It can remain dissolved in the iron matrix, as in austenite or martensite, or become part of a separate carbide crystal. Heat treatment helps determine how carbon is distributed between these states. The next chapters explain these transformations in detail.

More carbon can support greater hardness and wear resistance, but only within the balance of the full alloy and heat treatment. Too much carbide volume or an unfavourable matrix can reduce toughness, slow sharpening and, in stainless steels, consume chromium in chromium-rich carbides.

The carbon-chromium interaction is especially important. Carbon supports hardness; chromium supports corrosion resistance. More carbon can bind more chromium in carbides, leaving less available for passivation. A high-carbon stainless knife steel therefore needs enough chromium and a suitable heat treat-

ment to preserve strong corrosion resistance.

9.3 Chromium: the basis of stainless behaviour

Chromium makes stainless steel possible. The widely used compositional threshold is **at least 10.5% chromium by mass**. This classifies the steel; it does not mean every steel above the threshold resists corrosion equally well.

Chromium works through **passivation**, the formation of a very thin protective chromium-rich oxide film at the surface. This film helps separate the iron below from the environment. If scratched, it normally forms again when enough chromium and oxygen are available.

This explains both the value and the limit of the word *stainless*. The passive film can make a blade tolerate ordinary kitchen moisture very well, but salt, prolonged wetness, acids and contamination can still damage it.

Nominal chromium content also does not show how much chromium remains available for passivation. Chromium tied up in carbides cannot contribute fully to the protective film. As carbon rises, more chromium may be consumed in carbides. Two steels with the same nominal chromium can therefore corrode differently.

Heat treatment changes the balance again. During **austenitizing**, the high-temperature heating stage before quenching, some chromium carbides can dissolve and return chromium and carbon to the surrounding steel. Temperature, time and cooling affect both hardness and the chromium available in the final structure. Molybdenum and nitrogen can further improve resistance to localized corrosion.

Corrosion resistance comes from the whole system: chromium, carbon, chromium in solution, carbides, molybdenum, nitrogen, heat treatment, surface finish and environment.

The history of stainless cutlery illustrates the chemistry. In 1913 Harry Brearley produced a Sheffield chromium steel containing about 12.8% chromium and 0.24% carbon that became closely associated with the practical birth of stainless cutlery. Other researchers had already studied corrosion-resistant iron-chromium alloys, but Brearley's work helped turn the principle into an everyday advantage. The alloy was not immune to corrosion; it was easier to maintain because its surface could form a stable passive film.

Chromium can also form hard carbides that add wear resistance. But a chromium atom tied up in a carbide cannot simultaneously remain available in

the matrix for passivation. This competition is a clear example of alloy design as a balance.

9.4 Molybdenum: reliable hardening and pitting resistance

Molybdenum performs several useful supporting roles. It improves **hardenability**, which means the steel can form the intended hard structure more reliably during cooling after the high-temperature hardening stage. It also supports strength and stability at elevated temperatures.

In stainless steels, molybdenum improves resistance to localized corrosion such as **pitting**, where corrosion attacks small points and creates tiny pits in the surface. It reinforces chromium's protection rather than replacing it.

This helps explain why steels with similar chromium content can behave differently around salt and moisture. Molybdenum can also affect the heat-treatment window and, in some alloys, wear behaviour. The trade-offs are higher material cost and alloy complexity.

9.5 Vanadium: fine structure and very hard carbides

Vanadium's role depends on its quantity and the rest of the alloy.

At modest levels, it can help control the growth of the microscopic crystal grains during heat treatment. Keeping those grains fine can support toughness and edge stability. At higher effective carbide volume, vanadium forms extremely hard carbides that can raise abrasive wear resistance sharply.

High-vanadium powder steels can keep a slicing edge for a long time. The same carbides resist ordinary abrasives, so reprofiling or polishing may be slow on conventional stones. Diamond and other high-performance abrasives are often more efficient.

In knife steel, vanadium matters mainly for microstructural control and hard carbide formation, not rust resistance.

9.6 Tungsten: durable carbides and high-temperature stability

Tungsten forms hard carbides and increases wear resistance. It is common in tool steels and traditional Japanese blue-paper steels, where it supports a hard, durable cutting structure.

In a kitchen knife, tungsten can help maintain an aggressive edge for a long time. The same wear resistance also resists the sharpening stone, so sharpening may take longer or require more efficient abrasives. Very high hardness combined with thin geometry demands careful technique.

Tungsten is one part of the recipe, not a guarantee of superiority.

9.7 Cobalt: supporting demanding heat treatment

Cobalt is sometimes presented as if its presence automatically makes a blade stronger or more prestigious. Its role is more specific.

In suitable high-alloy steels, cobalt can support hardness and stability during high-temperature heat treatment, giving the heat treater more room to pursue demanding combinations of hardness and microstructure.

Cobalt alone does not make a knife tougher, more corrosion-resistant or easier to sharpen. Those properties depend on the complete alloy and process. It is expensive, but cost alone does not prove performance.

9.8 Nickel: toughness and visual contrast

Nickel can improve toughness in suitable steels and is widely used in pattern-welded Damascus because nickel-rich layers remain bright after etching and contrast with darker layers.

Etching means exposing the steel to a controlled chemical solution that attacks different layers at different rates. The different reactions make the Damascus pattern easier to see. Nickel-rich layers usually resist the etchant more strongly and remain brighter.

In many Damascus-clad knives, the patterned layers sit outside a separate core steel that forms the cutting edge. In full pattern-welded constructions, the

effect of nickel-bearing material depends on whether it reaches the edge and on the properties of the whole combination.

9.9 Nitrogen: another route to hardness and corrosion resistance

Nitrogen can perform some roles associated with carbon while leaving more chromium available for corrosion resistance. In suitable steels it contributes to hardness and can form **nitrides**, hard compounds created when nitrogen bonds with certain metals. Nitrides are conceptually similar to carbides, but they contain nitrogen instead of carbon.

This can combine fine edge behaviour with strong rust resistance. Sandvik 14C28N, used by Xinzuo, is an accessible example of a knife steel designed around a controlled carbon-nitrogen balance rather than simply maximizing carbon.

Nitrogen-alloyed steels require controlled industrial steelmaking, and their final performance still depends on heat treatment and geometry.

9.10 Why percentages are not enough

A composition chart cannot predict the complete performance of a finished knife.

The same nominal steel can be heat-treated to different hardness levels. One maker may favour toughness; another may push hardness for greater edge stability and wear resistance. Geometry also varies: one blade may be very thin behind the edge while another uses a more conservative cross-section. Surface finish, retained austenite, carbide distribution and manufacturing quality change corrosion resistance, toughness and sharpening response further.

Where an element ends up also matters. Chromium dissolved in the matrix can support passivation; chromium locked in carbides behaves differently. Vanadium in hard carbides adds wear resistance. A bulk percentage tells us what went into the steel, not exactly what exists in the final microstructure.

Read familiar marketing claims with the same care. “High carbon” points toward hardenability and potential hardness. “High chromium” supports corrosion resistance only when enough chromium remains in solution. “Vanadium carbides” point toward wear resistance. “Cobalt-added” points toward heat-treatment stability. “Nickel Damascus” often describes visual contrast rather

than an inherently better edge.

The conclusion is simple: **an alloy name describes ingredients and potential. The finished knife shows how well composition, heat treatment, microstructure and geometry were balanced.**

Chapter 10

Conventional and Powder-Metallurgy Steels

Composition is only the beginning. Once the alloy recipe is chosen, the next question is **how that chemistry becomes solid steel**.

Two steels with very similar compositions can develop different internal structures because they solidified in different ways. This matters especially in highly alloyed knife steels, where local carbide size and distribution can affect sharpening, toughness and wear resistance.

The sequence is simple: composition determines what is available; steelmaking affects how evenly it is distributed; heat treatment develops the final working structure.

10.1 First, picture the inside of steel

Steel looks uniform to the eye, but it is made from many microscopic crystals called **grains**. Together they form the solid metal.

The continuous iron-rich part of that structure is called the **matrix**. Inside the matrix there can be harder crystals called **carbides**, formed when carbon combines with iron or carbide-forming alloying elements such as chromium, vanadium, tungsten or niobium.

Think of the matrix as the continuous body of the material and the carbides as hard particles held inside it. The comparison is simplified, but it is enough

for understanding why manufacturing route matters.

A hard matrix helps the cutting edge resist deformation. Hard carbides can improve wear resistance. Their **size, amount and distribution** also matter. Many small, evenly distributed carbides behave differently from a few large clusters even when the total chemical composition is the same.

The next chapter on heat treatment will explain how the matrix can change between structures such as austenite and martensite and how carbides can dissolve or form during heating and cooling. Here we only need to understand how the steelmaking route affects the starting distribution.

10.2 Why a large ingot can become uneven

In conventional ingot metallurgy, molten steel is poured into a relatively large mould and allowed to solidify. It does not freeze everywhere at the same instant. Cooling progresses through the mass.

As this happens, some alloying elements can become more concentrated in certain areas. This uneven chemical distribution is called **segregation**. Carbides can also have more time to grow while the large mass cools.

Later forging and rolling improve the structure greatly, and modern conventional steels can be extremely clean and consistent. Many excellent knife steels do not need powder metallurgy. The challenge becomes greater when the alloy contains large amounts of carbon and strong carbide-forming elements, because these compositions are more prone to coarse or uneven carbide structures.

10.3 Powder metallurgy: solidifying millions of tiny droplets

Powder metallurgy also starts with molten steel. The important difference is what happens next.

Instead of letting the whole melt freeze as one large ingot, high-pressure inert gas breaks the liquid metal into **millions of tiny droplets**. This process is called **gas atomization**. Each droplet is so small that it cools and solidifies very quickly.

Because the droplets freeze quickly, alloying elements have less time and distance to separate and carbides have less time to grow. Each tiny particle therefore begins with a much finer and more even structure than a comparable region in a large slowly solidifying ingot.

The powder is then sealed and consolidated under very high temperature and pressure, commonly by **hot isostatic pressing, or HIP**. *Isostatic* means pressure is applied from all directions. The powder particles bond into a fully dense solid block of steel, which can then be forged and rolled into normal steel stock.

No loose powder remains inside the finished blade. The final material is dense steel.

A useful image is:

Conventional metallurgy solidifies one large body of steel and then works it into shape. Powder metallurgy solidifies millions of tiny pieces first, then joins them into one dense body.

10.4 Why this matters at a knife edge

Powder metallurgy can give highly alloyed steel a **finer and more even distribution of carbides and alloying elements** than conventional ingot processing of the same composition.

This can improve several things at once. Smaller carbides are less disruptive to a very fine cutting edge. More even distribution reduces local areas that are unusually weak or unusually resistant to abrasion. A highly alloyed steel can therefore carry a large population of wear-resistant carbides while retaining better toughness than the same composition would usually achieve with a coarser conventional structure.

This does not mean every powder steel is better than every conventional steel. A simple, well-designed conventional steel can have a very fine structure, excellent toughness and excellent edge behaviour. Powder metallurgy becomes most useful when the alloy composition itself would otherwise encourage large or uneven carbides.

It also does not make later processing unimportant. Poor heat treatment can still waste the potential of excellent powder steel, and poor geometry can still make an excellent alloy cut badly.

Xinzuo steels such as SRS13, M390, ZDP-189 and 14Cr14MoVNB should be read in this context. Their performance does not come from the word **powder** alone. It comes from composition, powder-metallurgy processing, heat treatment and blade geometry working together.

Remember the sequence: **the steel recipe tells us what elements are present; the manufacturing route affects how evenly the matrix and**

carbides begin; heat treatment develops the final microstructure; and edge geometry determines how that structure meets the food.

Chapter 11

Heat Treatment and Carbides: Turning Composition into a Blade

A steel grade is a recipe, not finished performance. Composition tells us which elements are present and which internal structures are possible. **Heat treatment decides which structures develop in the blade and in what proportions.**

To understand that process, we first need a simple picture of what steel looks like inside.

11.1 What steel looks like under a microscope

A finished piece of steel looks like one solid material, but it is not normally one single crystal. It is **polycrystalline**: it is made from a very large number of microscopic crystals called **grains**.

Inside each grain, iron atoms are arranged in a repeating geometric pattern. That arrangement can change when the steel is heated or cooled. Different arrangements are called **phases**. A phase is simply a form of the material with a particular internal structure and composition.

The most useful phases to recognize in knife steel are:

- **ferrite**: a relatively soft iron-rich phase often present before the blade is finally hardened;
- **austenite**: a high-temperature phase that can dissolve much more carbon and alloying elements than ferrite;
- **martensite**: the very hard structure formed when suitable austenite is cooled quickly enough during hardening;
- **retained austenite**: austenite that remains after **quenching**, the rapid cooling stage used during hardening, instead of transforming into martensite.

A slowly cooled carbon steel can also form structures such as **pearlite**, a fine mixture of ferrite and the iron carbide cementite. These structures matter in steel processing, but a hardened kitchen-knife edge is usually designed around martensite that is later tempered.

Austenite and martensite are **not carbides**. They are forms of the iron-rich part of the steel.

11.2 Matrix, grains and hard particles

Metallurgists often use the word **matrix**. The matrix is the continuous part of the steel that surrounds and supports the other microscopic parts. In a finished hardened knife, that matrix is commonly based mainly on martensite that has been **tempered**, meaning reheated after hardening to reduce internal stress and adjust its properties. Some retained austenite may also remain.

Think of the steel as an extremely fine stone floor. The continuous floor is the matrix. The floor itself is divided into many microscopic crystal grains. Inside it are harder particles such as carbides.

This comparison is simplified, but it explains why the same chemical recipe can behave differently after different processing. The hardness and toughness of the matrix, the size of its grains, the amount of retained austenite and the type, size and distribution of carbides all influence the finished blade.

11.3 What carbides are

Carbides are hard crystalline compounds made when carbon bonds with iron or certain alloying elements. They form naturally from the chemistry of the steel during solidification and later heating and cooling. They are not foreign grit added to the metal.

Some carbides form while the steel is being produced. Some remain undissolved when the blade is heated for hardening. Others form later as new, often much smaller crystals during cooling or tempering. Heat treatment can therefore dissolve, preserve or create different carbide populations.

Carbides matter because they are usually much harder than the surrounding matrix. They can improve resistance to abrasive wear, but they also affect toughness and sharpening. More carbide is not automatically better. **Type, hardness, size, amount and distribution all matter.**

With that picture in mind, the heat-treatment sequence becomes much easier to follow.

11.4 1. Preparing the steel before hardening

Before final hardening, steel often goes through earlier heating and cooling steps that soften it and prepare its internal structure for the final hardening cycle.

The basic purpose is to put the steel into a controlled starting condition. In many knife steels this means making it soft enough to machine or grind while keeping carbon and carbides distributed in a form suitable for the later hardening cycle.

Final heat treatment does not start from a blank state. Grain size, carbide size and distribution, chemical segregation and previous heating all affect what happens next.

11.5 2. Austenitizing: creating the high-temperature phase

During **austenitizing**, the steel is heated into a temperature range where austenite forms. Austenite has a crystal arrangement that can hold much more carbon in solution than ferrite.

As the steel heats, some existing carbides dissolve. Their carbon and alloying elements enter the austenite. Other carbides remain as solid particles.

The goal is not to dissolve every carbide. The heat treater wants the right amount of carbon and alloying elements in the austenite while preserving a useful carbide population and a fine grain structure.

Too little temperature or time can leave too little carbon or alloy in solution and limit the final hardness. Too much can dissolve unwanted carbide, grow the austenite grains and leave so much carbon and alloy in solution that excessive retained austenite becomes harder to control.

The correct austenitizing temperature and time therefore depend on the steel and the intended balance of hardness, toughness, corrosion resistance and wear resistance.

11.5.1 Vacuum heat treatment: controlling the atmosphere

At austenitizing temperatures, exposed steel can react strongly with air. It can oxidize and can lose carbon at the surface, a process called **decarburization**.

A **vacuum furnace** or another controlled protective atmosphere reduces these reactions. For knife manufacturing this means less scale, less surface carbon loss, cleaner parts and more controlled heating. It is especially useful for high-alloy stainless and tool steels that need high hardening temperatures.

Vacuum treatment does not itself make the steel hard. It controls the environment while the real heat-treatment stages take place.

11.6 3. Quenching: transforming austenite into martensite

After austenitizing, the blade is cooled quickly enough to prevent the steel from changing into the softer structures that would form during slower cooling.

When the cooling is fast enough, much of the austenite transforms into **martensite**. The iron atoms change arrangement before the carbon atoms have enough time to move to more comfortable positions. Carbon becomes trapped in a strained crystal structure.

That distortion is a major reason fresh martensite is so hard. It strongly resists permanent deformation, which is useful at a cutting edge.

The required cooling rate depends on the alloy. Some simple carbon steels need very fast cooling. Highly alloyed stainless and tool steels can harden with slower cooling because their alloying elements make the austenite more resistant to other transformations. This ability is called **hardenability**.

Fresh martensite is very hard but also highly stressed and relatively brittle. It is not normally the final condition wanted in a kitchen knife.

11.7 4. Retained austenite and sub-zero treatment

Not all austenite necessarily becomes martensite when the blade reaches room temperature. The part that remains is called **retained austenite**.

A controlled amount can be acceptable or even useful, but too much can make the blade's size and properties less stable over time and change how the edge responds in use.

Cooling the blade below room temperature can transform more retained austenite into martensite. Commercial terms such as **sub-zero**, **deep-freeze** and **cryogenic** are used for these low-temperature treatments, although the exact temperature ranges are not perfectly standardized.

A domestic freezer, dry ice and liquid nitrogen are therefore not equivalent processes.

For steels and heat-treatment recipes that benefit from it, sufficiently low-temperature treatment can:

- reduce retained austenite;
- increase hardness immediately after quenching;
- improve stability of size and shape;
- make the later tempering response more predictable.

It is not a universal upgrade. Converting more austenite into martensite also changes the hardness-toughness balance, so the treatment must suit the steel and the complete heat-treatment sequence.

11.7.1 Xinzuo as an example

Selected Xinzuo series specify **vacuum heat treatment and deep freeze**. These are different steps.

Vacuum processing controls the atmosphere during high-temperature treatment. Deep freezing is a later low-temperature step used when the steel and recipe call for further transformation of retained austenite.

Neither term should be read as a stand-alone quality badge. Its value comes from how it fits the complete treatment.

11.8 5. Tempering: making martensite usable

After quenching—and any specified sub-zero step—the blade is **tempered** by reheating it to a lower temperature.

Tempering reduces the internal stress of fresh martensite and adjusts the balance between hardness and toughness. It is more than simply making the steel softer.

During tempering, some trapped carbon can leave the martensite and form very small carbide crystals. The martensitic matrix becomes more stable. Retained austenite can also change, and some highly alloyed steels can form additional alloy carbides that increase hardness again. This is called **secondary hardening**.

Some steels use two or more tempering cycles. For example, retained austenite may become unstable during one temper, transform into fresh martensite during cooling and then need another temper so that new martensite is not left untempered.

The final target is not the highest possible HRC. It is the hardness that gives the required edge stability without sacrificing too much toughness or creating other problems for that steel and geometry.

11.9 6. Grinding after heat treatment

A correct furnace cycle can still be damaged later.

Grinding and sharpening generate heat at the surface. The edge is extremely thin, so it can overheat much faster than the thick body of the blade. If the hardened edge becomes too hot, it can be unintentionally tempered above the intended temperature and lose hardness or stability.

Controlled grinding, cooling and careful sharpening are therefore part of the heat-treatment result. Metallurgy does not stop when the blade leaves the furnace.

11.10 The main carbide families in knife steels

Carbides are crystals with different chemical compositions, and different carbide families have very different hardness and behaviour.

Metallurgical names such as **M7C3**, **M23C6**, **M6C** and **MC** are formulas. The letter **M** means one or more metals. For example, a chromium-rich M23C6

carbide can also contain iron, molybdenum or other metallic elements in some of its positions.

Carbide family	Typical forming elements in knife steels	Practical importance
Cementite, Fe₃C	Iron + carbon	Common in carbon and low-alloy steels; adds wear resistance but is much softer than vanadium or niobium carbides
Chromium-rich M₇C₃ / M₂₃C₆	Mainly chromium + carbon, often with other metals mixed in	Add wear resistance; chromium locked in these carbides is not available for stainless passivation
Vanadium-rich MC	Vanadium + carbon	Extremely hard; strongly increases abrasive wear resistance and can make ordinary abrasives work slowly
Niobium-rich MC	Niobium + carbon	Extremely hard; supports wear resistance and can help keep carbide structures fine in suitable alloys
M₆C and related Mo/W carbides	Molybdenum and/or tungsten with carbon and other metals	Important in many high-alloy tool-steel systems; contribute to wear resistance and secondary hardening
Fine tempering carbides	Composition depends on the steel and tempering temperature	Very small carbides that form during tempering and strengthen or stabilize the martensitic matrix

These are families, not a promise that every steel containing an element forms a separate pure carbide of that element. An element can remain dissolved in the matrix, join another carbide family or form a different phase depending on

composition and heat treatment.

11.11 Why carbide hardness changes sharpening

A sharpening stone must remove material from both the matrix and the carbide crystals inside it.

Aluminium-oxide stones can cut many common steel matrices and chromium-rich carbides effectively. Vanadium- and niobium-rich MC carbides are much harder. When a steel contains a large volume of them, diamond or another very hard abrasive may remove material more efficiently.

This is why two steels at the same HRC can feel very different on a sharpening stone. HRC mainly describes the bulk hardness of the steel; it does not tell us how hard every microscopic carbide is.

11.12 Why carbide size and amount affect toughness

Carbides are hard but relatively brittle. They interrupt the continuous matrix and create boundaries where stress can concentrate.

A small, well-distributed carbide population can give useful wear resistance with limited damage to toughness. Very large carbides or a very high carbide volume can make cracking easier, especially in a thin edge.

This is one reason powder metallurgy is valuable for many highly alloyed steels: it can produce smaller and more evenly distributed carbides than conventional ingot processing of the same composition. It does not remove the trade-off completely. A powder steel with a very high carbide fraction can still be less tough than a lower-carbide steel.

11.13 Why heat treatment changes the carbide population

The carbide structure of a finished knife is not identical to the one that existed before hardening.

During austenitizing, some carbides dissolve and release carbon and alloying elements into austenite. During cooling and tempering, new carbides can form.

The final blade therefore contains a result created by both **composition and its complete heating and cooling history**.

This redistribution changes several properties at the same time:

- carbon dissolved before quenching helps produce hard martensite;
- chromium and molybdenum left in the matrix can support corrosion resistance;
- hard carbides improve abrasive wear resistance;
- carbide size and volume affect toughness and sharpening response;
- grain size and retained austenite further change the behaviour of the matrix.

The important sequence is:

Composition defines what can form. Heat treatment creates the final matrix and carbide population. Geometry and sharpening determine how that microstructure works at the cutting edge.

Chapter 12

Xinzuo Blade Steels: Different Balances for Different Knives

Xinzuo uses many blade steels because different knives, users and tasks need different balances of hardness, edge retention, sharpening response, corrosion resistance and toughness. This chapter explains the practical role of each material in the current range.

No steel maximizes every desirable property. A fine slicer, general chef's knife and heavy chopper need different balances. The useful question is not which steel ranks highest, but which steel best matches the knife's geometry, heat treatment, work and maintenance needs.

A steel name describes potential. Heat treatment, blade geometry, thickness behind the edge, sharpening and use determine how that potential appears in the finished knife. Heat Treatment and Carbides explains the manufacturing side; this chapter focuses on the practical position of each steel.

12.1 A compact comparison

Steel	Origin / source	Published hardness	What makes it distinctive in practice	Main compromise or limitation
ZDP-189	Japan	65-67 HRC	Extreme hardness and very high wear resistance give excep- tionally long slicing-edge life when geometry and use are appropriate	Demanding sharpening; thin edges require disciplined use and should not be twisted or impacted
SRS13 powder steel	Japan	62-64 HRC	Powder- metallurgy structure gives a strong combination of fine cutting response, stainless care and high retention without pushing hardness as far as ZDP-189	More wear- resistant and less forgiving to sharpen than simpler stainless or carbon steels

Steel	Origin / source	Published hardness	What makes it distinctive in practice	Main compromise or limitation
Böhler M390	Austria, Böhler	60-62 HRC	Premium powder- metallurgy stainless with very high wear resistance and strong corrosion resistance; excellent when long edge life and stainless care matter together	Independent knife-scale testing places its toughness relatively low compared with tougher knife steels; efficient abrasives are useful for sharpening
Blue Steel No. 2	Japan	62-64 HRC	Reactive high-carbon steel that can take a very keen edge and respond unusually well to water stones despite high hardness	Requires immediate cleaning and drying and will naturally develop patina

Steel	Origin / source	Published hardness	What makes it distinctive in practice	Main compromise or limitation
14Cr14MoVNb powder steel	Xinzuo	60-64 HRC	Niobium- bearing premium powder steel intended to combine high hardness and retention with a finer, more controlled carbide structure	Premium sharpening require- ments; exact behaviour depends strongly on heat treatment and geometry
12Cr18MoV	Xinzuo	59-61 HRC	Xinzuo's newer all-round conventional stainless formulation, designed for a particularly strong combination of corrosion stability, useful hardness, toughness and straight- forward maintenance	Less extreme wear resistance than the hardest powder steels, by design

Steel	Origin / source	Published hardness	What makes it distinctive in practice	Main compromise or limitation
AUS-10	Japan, Aichi family	58-62 HRC	Japanese high-carbon stainless with a practical balance close to the performance class commonly associated with VG10: good sharpness, hardness, corrosion resistance and manageable maintenance	Moderate wear resistance compared with the premium powder-steel tier; not chemically identical to VG10

Steel	Origin / source	Published hardness	What makes it distinctive in practice	Main compromise or limitation
10Cr15CoMoV	China / Xinzuo specification	58-62 HRC	Xinzuo's long- standing high-carbon stainless core, tuned for a mature balance of edge retention, toughness, hardness and sharpening response, with especially strong corrosion stability in Xinzuo's implementa- tion	Does not maximize one single property; final performance remains highly heat- treatment- and geometry- dependent

Steel	Origin / source	Published hardness	What makes it distinctive in practice	Main compromise or limitation
VG10	Japan, Takefu Special Steel	HRC 60 or higher in Takefu’s published specification	Established Japanese market benchmark for balanced high-carbon stainless perfor- mance: good corrosion resistance, durable sharpness, wear resistance and manageable sharpening	Balanced rather than extreme; mature market standard whose finished behaviour still depends strongly on heat treatment, geometry and sharpening
Sandvik 14C28N	Sweden	57-59 HRC	Nitrogen- assisted stainless design with fine structure, strong corrosion resistance and very good edge stability; pleasant for users who maintain their own knives	Moderate wear resistance and edge life compared with high-carbide powder steels

Steel	Origin / source	Published hardness	What makes it distinctive in practice	Main compromise or limitation
440C	International standard grade; Xinzuo supply from China	56-58 HRC	Proven high-carbon martensitic stainless with useful wear resistance, strength and corrosion resistance at an accessible cost	Coarser carbide structure and more modest retention than newer high- performance steels at Xinzuo's published hardness
DIN 1.4116	Germany	56-58 HRC	Tough, corrosion- resistant and forgiving in everyday use, with a large margin against chipping and misuse compared with very hard thin-edge steels	Needs more frequent edge maintenance and can form a persistent burr; moderate hardness does not au- tomatically mean effortless sharpening

This table shows compromises, not rankings. Two steels can feel almost identical in ordinary kitchen use yet separate clearly in a laboratory wear or corrosion test. Two steels at similar hardness can also feel completely different on a sharpening stone.

12.2 Premium powder-metallurgy steels

12.2.1 ZDP-189: maximum hardness and long edge life

ZDP-189 sits at the extreme-hardness end of the range. At 65-67 HRC, it strongly resists edge deformation and can keep cutting for a long time when geometry and use are appropriate. Its performance comes from hardness and wear resistance together, not HRC alone.

The same properties narrow its margin for misuse. High hardness and a large carbide population slow sharpening, while a thin edge tolerates lateral loads poorly. ZDP-189 is for controlled cutting, not twisting through hard food, levering, striking bone or general abuse.

12.2.2 SRS13: premium performance without ZDP-189's extreme hardness

SRS13 is specified at 62-64 HRC and used as a high-performance Japanese powder-steel core. Powder metallurgy distributes alloying elements and carbides more evenly than conventional high-alloy ingot processing, supporting a refined edge and strong wear resistance.

SRS13 remains a premium high-retention steel without reaching ZDP-189's hardness. It suits users who want a fine stainless edge and long cutting life without going to the hardest steel in the range. Sharpening still takes more time than with simpler steels, so good abrasives and controlled technique help.

12.2.3 Böhler M390: wear and corrosion resistance first

M390 is a premium Austrian powder-metallurgy martensitic stainless steel from Böhler. Its main strength is **very high wear resistance combined with strong corrosion resistance**. It suits knives that need long working-edge life with stainless convenience.

Its toughness is more modest. Böhler describes both its toughness and **ductility—the ability to deform before breaking—as good**, while independent knife-scale impact testing places optimized M390 toward the lower-toughness side of the knife-steel spectrum. Earlier unusually high toughness results were linked to heat treatment with excessive retained austenite. This does not make M390 fragile in normal kitchen use; it means its strongest reasons for selection are wear resistance, corrosion resistance and retention rather than exceptional toughness.

Its carbide population also resists ordinary abrasives. Hard ceramic, diamond

or **cubic boron nitride (CBN)** abrasives can make sharpening much more efficient. CBN is an extremely hard synthetic abrasive used for difficult-to-abrade steels.

12.2.4 14Cr14MoVNb: Xinzuo's niobium-bearing premium powder steel

Xinzuo's 14Cr14MoVNb is a niobium-bearing powder steel used in premium layered constructions. Niobium is a strong carbide former and, in a suitable alloy, can create very hard MC-type carbides while supporting microstructural refinement. The goal is to balance hardness, wear resistance, carbide size and edge stability, not simply add a prestigious element.

The steel is positioned as a premium high-retention option with a broader balance than a maximum-hardness strategy. Selected series use **vacuum heat treatment and deep-freeze processing**, which help determine how much of the alloy's potential reaches the finished blade.

Niobium alone cannot predict performance. Composition and heat treatment still determine the final carbide population.

12.3 Carbon and all-round high-performance stainless steels

12.3.1 Blue Steel No. 2: high hardness, responsive sharpening

Blue Steel No. 2, or Aogami #2, is the main reactive carbon-steel option in the range and is specified at 62-64 HRC. It reaches high performance through a simpler route than M390 or ZDP-189. Its high-carbon alloy favours a keen, stable edge without relying on a large volume of corrosion-resistant alloy carbides.

A well heat-treated Blue #2 blade can respond very quickly to suitable water stones despite its hardness. It can reach the apex and deburr cleanly, which is why many experienced sharpeners find it more pleasant than lower-HRC stainless steels with greater abrasive wear resistance.

Its main demand is corrosion care. Wash and dry it immediately. Expect a **patina**, the stable grey, blue or dark surface reaction that develops naturally on reactive carbon steel, and distinguish it from active orange rust.

12.3.2 10Cr15CoMoV: the long-standing Xinzuo all-round reference

10Cr15CoMoV has long been one of Xinzuo's characteristic steels and, for years, its most widely used high-carbon stainless core. Xinzuo selected and heat-treats it for a broad balance of edge retention, toughness, corrosion resistance and sharpening practicality rather than one extreme property.

That balance gives it an exceptional **quality-to-price ratio**. It can support thin, refined kitchen edges and serious professional use without the material cost or sharpening demands of premium powder steel. Hardness, toughness and sharpening response remain in the familiar high-performance all-round stainless range, so the steel is predictable both in use and on the stone.

One of the strongest parts of Xinzuo's implementation is corrosion behaviour. Internal comparative testing shows a particularly stable response to corrosion for this steel and heat-treatment combination. This gives the user a generous practical margin around moisture, food acids and everyday kitchen exposure without giving up the edge stability or maintenance characteristics expected from a high-carbon stainless core.

Its value lies in how completely these properties are balanced. It is not intended to win one isolated laboratory category; it is intended to leave very few weak points in normal kitchen use.

12.3.3 12Cr18MoV: the newer everyday-use formulation

12Cr18MoV is Xinzuo's newer all-round stainless formulation, used as a core in conventional and copper-Damascus constructions. Current product-development information dates its introduction to around **2024**, making it the newest conventional stainless core steel discussed here.

It pushes the same everyday-use philosophy further. Xinzuo's formulation emphasizes **very strong corrosion stability together with useful hardness, edge stability, practical toughness and straightforward maintenance**. The result is a conventional stainless steel that remains easy to live with while offering an unusually complete performance envelope for daily kitchen work.

Xinzuo currently rates 12Cr18MoV among its strongest everyday choices. It sits above the older conventional all-round tier in the company's internal product positioning, not because it maximizes a single headline number, but because the balance of corrosion resistance, cutting stability, toughness and maintenance is stronger as a whole.

It still targets different priorities from M390, ZDP-189 or Blue #2. Those steels pursue more specialized extremes; 12Cr18MoV is designed to make the all-round stainless formula as complete as possible without the sharpening demands of the hardest powder steels.

12.3.4 AUS-10: a Japanese all-rounder near the VG10 class

AUS-10 is a Japanese stainless cutlery steel from the Aichi family. Aichi lists it among its higher-carbon knife steels, with molybdenum and vanadium added to improve hardness, wear resistance and tempering resistance. Xinzuo commonly specifies it at 58-62 HRC.

For kitchen use, AUS-10 sits in much the same practical class as **VG10**. They are chemically different, but both are Japanese high-carbon martensitic stainless steels designed to combine good sharpness, useful hardness, corrosion resistance and manageable maintenance.

AUS-10 offers more performance than basic stainless steels while remaining easier to maintain than the highest-wear powder steels. Its strength is balance rather than dominance in one metric.

12.3.5 VG10: the Japanese benchmark and why it matters to Xinzuo

VG10 is a proprietary Japanese stainless cutlery steel produced by **Takefu Special Steel**. Takefu publishes about 1% carbon, 15% chromium, 1% molybdenum, 0.25% vanadium and roughly 1.55% cobalt, with quenched hardness of HRC 60 or higher. The manufacturer highlights corrosion resistance, durable sharpness, wear resistance, ease of manufacturing and relatively straightforward sharpening.

VG10 became one of the defining reference steels for premium mass-market Japanese kitchen knives. Its importance is therefore partly historical and commercial: it established a familiar expectation of roughly 60 HRC, good stainless behaviour, useful edge life and manageable sharpening in one package.

That makes VG10 a valuable **benchmark**, not an upper limit for the category. Modern or differently optimized all-round stainless formulations can preserve the same broad virtues while moving more performance toward corrosion stability, edge behaviour or another part of the balance. The steel name alone also cannot show the quality of the heat treatment, geometry or finished blade.

At the time of this edition, **VG10 is not a normal Xinzuo production steel because the producer restricts supply for material destined for China.** Current Xinzuo procurement information indicates that Takefu does not normally authorize VG10 supply for Chinese production. Xinzuo is discussing specific authorization to purchase and use genuine VG10. This may change in future editions and is a procurement issue, not a metallurgical property.

Verified supply matters because Takefu publicly warns about counterfeit or similar products sold under the VG10 name. If Xinzuo adopts it, genuine Takefu material matters more than the letters printed on the blade.

12.3.6 Sandvik 14C28N: fine structure, nitrogen and edge stability

Sandvik developed 14C28N specifically with knife applications in mind. Nitrogen helps support corrosion resistance while maintaining a fine structure and good edge stability.

14C28N favours a clean, stable edge, strong stainless behaviour and straightforward maintenance rather than M390-level carbide wear resistance. It is especially attractive to users who sharpen their own knives and prefer predictable sharpening to the longest possible interval between sessions.

Its moderate published hardness does not prevent extreme sharpness. Geometry and sharpening create sharpness; steel affects how the edge survives, fails and is restored.

12.4 More forgiving stainless steels

12.4.1 440C: the classic high-carbon stainless work steel

440C is a classic high-carbon martensitic stainless steel with a long history in knives, bearings and other wear-resistant components. It provides more carbon and wear resistance than simpler low-carbon stainless grades while retaining good corrosion resistance.

Its carbide structure is generally coarser and its performance ceiling lower than newer powder steels, but it remains a useful working material. At Xinzuo's published 56-58 HRC, it offers useful strength, decent edge life, good stainless care and accessible cost.

Think of it as a proven work steel rather than a prestige steel. It trades some

fine-edge retention and high-performance potential for accessibility and predictable service.

12.4.2 DIN 1.4116: where its forgiveness comes from

Xinzuo specifies DIN 1.4116 at 56-58 HRC in series such as B13S and B35. It emphasizes **corrosion resistance, toughness and forgiving everyday use**. It suits kitchens with less disciplined technique, frequent washing or users who value robustness more than maximum retention.

Moderate hardness can make material removal easier than on a very wear-resistant powder steel, but it does not guarantee easy sharpening. Carbide structure, heat treatment, geometry and burr behaviour also matter. DIN 1.4116 can form a persistent burr and may feel less clean on a water stone than the harder but comparatively fine Blue Steel No. 2.

A better description is **forgiving in use and generally maintainable with ordinary abrasives, but not automatically the easiest steel to apex and deburr**. It usually needs maintenance sooner than harder high-retention steels but offers more margin against chipping and misuse.

Hardness alone is therefore a poor shortcut for sharpening behaviour.

12.5 Layered constructions are not separate “sharpness levels”

Xinzuo also uses layered constructions including conventional Damascus cladding, copper-Damascus designs and 110-layer full Damascus.

Layer count does not directly measure sharpness. In a core-and-cladding blade, the core steel, its heat treatment and edge geometry control the apex. Decorative copper or brass layers can create strong contrast without making the edge inherently sharper.

Full-Damascus constructions differ because alternating materials can continue through the edge and affect wear there. The Self-Sharpening Effect in Full Damascus Blades explains this differential wear. It does not mean normal sharpening becomes unnecessary.

12.6 Reading the Xinzuo steel range

The range uses several metallurgical strategies. ZDP-189, SRS13, M390 and 14Cr14MoVNB pursue premium retention through high hardness, powder metallurgy or high-carbide performance, with different trade-offs. Blue Steel No. 2 reaches high hardness through a simpler reactive-carbon route and remains highly responsive on water stones. VG10 remains a useful external benchmark for the established all-round high-carbon stainless class. Xinzuo's 10Cr15CoMoV represents a mature in-house balance with particularly strong corrosion stability, while 12Cr18MoV pushes the company's conventional everyday-use balance further. AUS-10 occupies the same broad all-round stainless centre. 14C28N, 440C and DIN 1.4116 move toward easier care, greater forgiveness and more accessible maintenance.

Do not buy the steel at the top of a table. Choose the Xinzuo knife whose steel, heat treatment, geometry, maintenance needs and sharpening response fit the work and the user.

Source note: Xinzuo steel names, published hardness ranges, series use and company-specific positioning are based on Xinzuo product information and current internal product-development information. The specification of the individual model takes precedence. M390 property descriptions were cross-checked against Böhler M390 MICROCLEAN technical data and independent knife-steel toughness testing by Larrin Thomas. AUS-10 origin and intended blade use were cross-checked against Aichi Steel's cutlery-steel information. VG10 composition and manufacturer properties were cross-checked against Takefu Special Steel's published VG10 data. Statements about the approximate 2024 introduction and internal positioning of 12Cr18MoV, Xinzuo's 10Cr15CoMoV corrosion testing, and current VG10 procurement discussions are Xinzuo internal information rather than claims published by the steel producers.

Chapter 13

Xinzuo Handle Materials - Feel, Care and Choice

The blade gets most of the attention, but the handle also changes how a knife feels and works. Its material affects weight and balance, grip with dry or wet hands, resistance to water and cleaners, durability, maintenance and appearance. Shape and finish still determine much of the actual grip, so always judge the material as part of the complete handle.

No material is universally best. Synthetic materials can resist moisture and need little care. Natural wood feels warmer and gives every handle a unique appearance, but it needs more attention. Engineered materials combine fibres, wood or fabric with resin to balance these qualities.

A **composite** is simply a material made by combining two or more different materials so they work together. A **laminated** is a composite built from bonded layers. A **thermoplastic** is a plastic that can be softened with heat and shaped, then becomes firm again as it cools. These terms appear often in handle descriptions.

For practical purposes, this chapter uses three groups: **synthetic and technical composite materials**, **semi-synthetic or hybrid materials**, and **natural materials**. This is a teaching classification, not a strict scientific classification. G10, carbon fibre and Micarta, for example, are technically all composites.

Xinzuo's documented range includes rosewood, ebony, North American desert ironwood, red sandalwood, resin, red wood, figured sycamore, olivewood, oak,

carbon fibre, Pakka wood, Micarta, carbon fibre with resin, walnut and black G10. Selected series also use copper, buffalo horn, imitation ox bone and other accents. Polypropylene and ABS appear below only as useful industry comparisons because they are common on lower-cost and highly functional mass-produced knives; they are not presented as documented Xinzuo handle materials.

Keep one principle in mind: **material changes the possibilities, but shape creates the grip**. A simple material shaped well for the user can be more comfortable than a rare material shaped poorly.

13.1 Read the material by design consequence

Instead of ranking materials in a list of advantages and disadvantages, ask what each family changes in the finished handle.

Material family	What it lets the designer control	What the customer should verify in hand
Injection-moulded thermoplastics such as PP and ABS	Low weight, repeatable shape, hygiene and economical production	Surface traction, seam finishing, heat exposure and long-term surface wear
Structural composites such as G10, Micarta and carbon fibre	Moisture stability, stiffness, controlled texture and engineered weight	Density, edge finishing, texture under prolonged use and balance with the blade
Wood-resin materials such as Pakka wood and stabilized wood	Natural visual character with lower moisture movement than untreated wood	Quality of impregnation or lamination, surface finish and assembly
Natural woods such as rosewood, ebony, ironwood, olivewood, oak and walnut	Individual grain, warm touch and a wide range of densities	Actual weight, dimensional stability, finish condition and willingness to provide wood care
Decorative resins and mixed constructions	Repeatable colour, visual depth, spacers and contrasting accents	Scratch visibility, transitions between materials and care required by the most sensitive component

The individual profiles below explain the materials used by Xinzuo. Compare them through weight, moisture response, surface grip, manufacturing method and care rather than through a universal best-to-worst order.

13.2 Synthetic and technical composite materials

Synthetic resins and technical plastics cover a broad family. They can form a handle directly or bind fibres and other reinforcement in a composite. Their main advantages are consistency, moisture resistance and repeatable manufacturing. Colour, texture and dimensions are easier to control than with natural wood.

The word **plastic** says little about quality. An inexpensive injection-moulded thermoplastic and a glass-fibre laminate such as G10 are both synthetic, but their structure, process, cost and mechanical behaviour differ greatly.

13.2.1 Polypropylene (PP)

Polypropylene is common on inexpensive knives and professional knives designed around practicality, hygiene and low maintenance. **Injection moulding** means forcing softened plastic into a mould so it takes the required shape. It produces light, water-resistant handles quickly and cheaply, with texture formed directly into the surface.

A good PP handle can last for years, but heavy use can scratch or polish the surface smooth. Long exposure to ultraviolet light from sunlight, high heat, aggressive chemicals and harsh repeated washing can also cause discolouration, deformation or make some formulations more brittle.

PP is not inherently poor. Its limitation on a high-end knife is that it offers less rigidity, tactile richness, shaping freedom and visual distinction than G10, carbon fibre or selected woods. Premium makers usually choose other materials when the handle is meant to communicate craftsmanship and product identity.

13.2.2 ABS

ABS, or acrylonitrile-butadiene-styrene, is another injection-mouldable thermoplastic. It combines useful rigidity, impact resistance, water resistance and low cost, and it can form complex shapes and textures consistently.

Its surface can scratch, while heat and chemical resistance depend on the formulation. ABS works well when cost, repeatability and fast production matter

most. High-end knives usually choose G10, carbon fibre, Pakka wood, selected woods or other engineered materials for greater finishing freedom, tactile refinement and visual distinction.

13.2.3 G10

G10 is made from layers of woven glass-fibre cloth impregnated with **epoxy resin**, a hard-setting resin used to bond and protect the fibres. The layers are compressed and cured under heat and pressure into a dense laminate.

It absorbs very little water, resists swelling and remains dimensionally stable, meaning its size and shape change very little with normal humidity. It also tolerates oils, grease, many detergents and ordinary temperature changes. Shaping and texturing can create a secure grip even with damp hands, making G10 well suited to frequent and demanding use.

Its compromises are mainly tactile and aesthetic. G10 feels more technical and less warm than wood and can be relatively dense. That weight can be useful when the design needs more mass toward the handle. Very aggressive texture can also become uncomfortable during long sessions, so surface geometry and edge finishing still matter.

13.2.4 Carbon fibre

A carbon-fibre handle usually uses fine carbon filaments woven or layered inside resin, often epoxy. The resin surrounds and holds the fibres together. Once cured, the composite can be very rigid without being heavy.

This allows rigid, strong handles at low weight. The reinforcement absorbs essentially no water, the composite changes very little with normal humidity and the woven surface gives a distinctive technical appearance. A light carbon-fibre handle can shift balance toward the blade, although the complete knife determines the final result.

The main disadvantage is cost. Raw material and shaping are more expensive than for many common handle materials. Highly polished surfaces can offer less traction than textured G10 and can show fingerprints or fine scratches. **Carbon fibre** also does not guarantee identical quality in every product; reinforcement, resin, fibre direction and manufacturing all matter.

13.2.5 Carbon fibre with resin

Combining carbon reinforcement with coloured or transparent resin adds visual freedom while keeping much of the stability of an engineered composite. Xinzuo

uses these constructions when the handle itself is a strong design feature.

Appearance does not reveal the amount or direction of carbon reinforcement. Product descriptions should distinguish structural carbon-fibre laminate from decorative carbon/resin composite only when the specification confirms it.

13.2.6 Resin

Resin handles and resin-rich components offer repeatable colour, good moisture stability and broad decorative freedom, including effects such as embedded gold leaf. They can reproduce a design more consistently than natural wood.

They need little maintenance but can still scratch, lose gloss under abrasion or suffer heat damage. Shape, polishing and surface texture determine much of the final feel.

13.3 Semi-synthetic and hybrid materials

These materials combine a natural or fibre-based component with resin. The goal is to keep some appearance, structure or tactile character of the original material while reducing moisture absorption and changes in size or shape.

13.3.1 Micarta

Micarta is made by bonding layers of fabric, canvas, paper or other fibre material with **thermosetting resin**, a resin that hardens permanently during curing and does not simply soften again like a thermoplastic. The stack is compressed and cured into a solid laminate. Linen and fabric versions are common in knife handles.

The result is tough, stable and far less sensitive to humidity than natural wood. Shaping the handle cuts through the layers and exposes lines and colour transitions that can resemble natural grain.

Surface finish controls much of the grip. Polished Micarta feels smoother; a finish that exposes more fabric structure gives more traction without necessarily feeling abrasive. Use can gradually create a slight patina.

George Westinghouse developed Micarta in 1910 as an electrical insulating laminate. Paper or fabric layers bonded with resin created a strong, stable industrial material, later used in lightweight aircraft structures during the First World War. Tool and knife makers adopted it much later because the same stability, toughness and ease of shaping suit handles. Micarta is therefore an engineering material, not simply an imitation of wood.

It generally feels warmer and less industrial than G10 while retaining many technical advantages. Quality still depends on the fibres, resin and manufacturing process.

13.3.2 Pakka wood

Pakka wood is a characteristic engineered handle material used by high-end knife makers, including Xinzuo. It is not a wood species. Thin layers of wood or wood-derived material are impregnated with resin, often coloured, and compressed into a dense block that is later cut and shaped into handles.

The process keeps part of wood's visual structure and warmth while improving resistance to moisture, abrasion, deformation and cracking. Xinzuo uses Pakka wood across several Damascus and three-layer composite families because it combines a wood-like appearance with greater day-to-day stability.

Well-made Pakka wood absorbs much less water than untreated wood and has a more stable, less porous surface. **It generally tolerates detergents and degreasers better than untreated natural wood**, while the instructions for the knife and cleaning product still take precedence.

It is not indestructible. Strong chemicals, prolonged soaking, high temperatures and dishwasher cycles can damage the resin, finish or handle assembly.

Pakka wood also gives the manufacturer strong control over appearance. Layers can be coloured before bonding, and shaping exposes regular decorative lines. It gives up some of the uniqueness of solid wood in exchange for lower maintenance and better humidity stability, which can be an excellent compromise for frequent use.

13.3.3 Stabilized wood

Stabilized wood starts with real natural wood, often chosen for attractive figure. It is commonly placed under vacuum so a very fluid resin can penetrate pores and internal cavities, then cured into a wood-and-resin composite.

This reduces moisture uptake and makes the wood less likely to swell or shrink. It can also make highly figured, porous or fragile pieces suitable for handles when untreated wood would be too unstable.

The material keeps much of the original wood's unique appearance while behaving more like a technical composite. The process adds weight, cost and complexity. Stabilized wood is not automatically waterproof; performance depends on the starting wood and how deeply and evenly the resin penetrates.

13.4 Natural materials

Natural wood remains popular because every piece has its own grain, tone and small variations. No two handles are perfectly identical, and wood has a warm tactile character that fully synthetic materials struggle to reproduce.

Wood is **hygroscopic**, which means it exchanges moisture with the surrounding air and water. Repeated water exposure, high humidity or excessive drying can make it expand or contract and, in severe cases, distort or crack. Dry wooden handles after washing and condition them only when the surface needs it, using a suitable food-contact-safe mineral oil or conditioner.

13.4.1 Rosewood

Rosewood is a commercial term covering several tropical woods rather than one botanical species. They are usually fairly dense and hard, with colours from reddish brown to very dark brown and clearly marked grain.

Xinzuo uses rosewood in several series. It takes a smooth finish well, gives each knife an individual appearance and usually resists wear better than softer woods.

It still requires natural-wood care: avoid soaking and condition it when needed. Its density can also add substantial handle weight and change balance.

13.4.2 Ebony

Ebony comes from several *Diospyros* species, mainly from tropical Africa and Asia. It is known for very dark colour, fine grain and high density.

These properties allow smooth, precise surfaces and suit clean geometric forms, including octagonal Asian handles. Xinzuo uses ebony with both Damascus and conventional stainless blades.

Its density gives good wear resistance and stability but also adds weight. Ebony still needs protection from soaking, repeated wet-dry cycles and excessive drying, and it is generally more expensive than common handle woods.

13.4.3 North American desert ironwood

Desert ironwood, usually *Olneya tesota*, comes from the Sonoran Desert of the southwestern United States and northwestern Mexico and is one of the densest premium woods used for knife handles.

It is extremely hard, dense and naturally rich in oils. This gives strong wear resistance and better dimensional stability than many common woods, along with deep contrasting figure and a refined polish.

Xinzuo uses it in high-end families including selected ZDP-189, SRS13 and full-Damascus knives. Its main compromises are cost, weight and difficult shaping. Despite its density and natural oils, it is not waterproof and should still be dried promptly.

13.4.4 Red sandalwood and red wood

These names describe red-toned woods used in selected Xinzuo handles. They provide warmth, traditional character and strong contrast with polished steel. Commercial wood names do not always identify one species clearly, so product descriptions should use the authorized series name rather than inventing an undocumented species or origin.

Care is simple: keep them out of the dishwasher, do not soak them, dry promptly and store away from strong heat.

13.4.5 Figured sycamore

Figured sycamore is chosen for the movement and depth of its grain. The surface can appear to change as light moves across it, making the handle a visible part of a premium design.

The figure is naturally variable and cannot be perfectly standardized. The wood also needs normal protection from prolonged moisture, dishwasher cycles and strong heat.

13.4.6 Olivewood

Olivewood comes from *Olea europaea* and is strongly associated with the Mediterranean. It is hard and compact, usually cream, honey or light brown with darker irregular veins. These variations make each handle visibly different.

It feels warm, takes a good polish and balances hardness, weight and visual character well. Xinzuo uses olivewood with several constructions, including 14Cr14MoVNb Damascus, 10Cr15CoMoV composite steel and 10Cr15CoMoV-core Damascus. Some model specifications identify Italian olivewood.

Olivewood is more sensitive to repeated wet-dry cycles than resin-impregnated materials. Dry it promptly and condition it when appropriate. Its strong

natural variation also means a product photograph cannot promise identical grain on another handle.

13.4.7 Oak and burnt oak

Oak has a bold, familiar grain. Burnt or darkened finishes increase contrast and give selected Xinzuo handles a more rustic or architectural appearance.

Its relatively open grain makes surface finishing important. A properly finished oak handle works well in normal use, but soaking and dishwasher cleaning can raise fibres, stain the surface or weaken the assembly over time.

13.4.8 Walnut

Walnut comes from several *Juglans* species. Its colour ranges from medium brown to deep dark tones, usually with elegant grain less dramatic than olive-wood.

It is generally lighter than ebony and desert ironwood, which can help avoid moving the balance too far toward the handle. It feels warm and is relatively easy to shape and finish.

Like other natural woods, walnut needs more care than G10, Micarta or Pakka wood. Prolonged water exposure and strong humidity changes can move the material. Normal use may also mark or age the surface; this should be distinguished from cracking, loose assembly or finish failure.

13.5 Accent and combination materials

Selected Xinzuo handles combine the main grip material with copper, white or black buffalo horn, imitation white ox bone, metal or composite spacers. These accents can change balance, reinforce transitions and create contrast.

Treat a combined handle according to its most sensitive material. For example, **olivewood with black buffalo horn** is one construction, not two independent care systems. Natural horn, when genuinely used, varies in colour and should be protected from strong heat and prolonged water exposure.

13.6 Material, ergonomics and balance

Judge handle material together with handle shape and the whole knife. Two handles with the same geometry can feel very different in carbon fibre and

ebony because the materials have different densities. Surface finish changes the experience again: stronger texture can improve grip, while a smoother surface can feel more comfortable during long use.

Handle material is not only an aesthetic choice. G10, Micarta, Pakka wood, carbon fibre and natural woods solve the same basic problem in different ways: creating a stable, comfortable and secure interface between hand and blade. The best choice depends on the knife's design, balance, working environment and user.

13.7 What a material name does not establish

The material name alone does not establish:

- the exact botanical species behind every trade name;
- the resin system or reinforcement percentage of every composite;
- sustainability certification or geographic origin unless separately documented;
- identical colour or grain from one knife to another;
- dishwasher suitability;
- identical balance across knives with different blade sizes.

These details require verified product data. Do not turn an attractive material label into an unsupported scientific or sourcing claim.

13.8 Choosing a handle by use

Rosewood, Olivewood, Walnut, Oak and Figured Sycamore emphasize natural grain and a warm feel. Ebony and Desert Ironwood feel denser, darker and more substantial. Pakka wood keeps a wood-like appearance with greater consistency and easier care, while stabilized wood preserves the individuality of real wood while reducing moisture sensitivity.

Micarta and G10 emphasize stability and durability in demanding or humid work. Carbon fibre reduces weight and gives a modern technical appearance, while carbon/resin and resin composites offer more design freedom and repeatable decorative effects. Polypropylene and ABS represent the other end of the engineering spectrum: highly practical and economical, but usually less distinctive and less flexible in premium finishing.

None of these materials can compensate for a handle that does not fit the user's hand or movement. Grip shape and size matter more than the material name alone.

13.9 Care by failure mode

- **Water:** hand-wash briefly with mild soap, then dry the blade, handle and joints. Do not leave the handle immersed or wet.
- **Heat and chemistry:** keep the knife out of the dishwasher and away from high heat or aggressive cleaners that can attack resin, adhesive, finish or natural material.
- **Abrasion:** use non-abrasive cleaning on polished resin, carbon fibre and fine wood surfaces.
- **Moisture movement:** inspect natural wood and condition it only when the surface actually becomes dry, using a suitable food-contact-safe product.
- **Assembly:** check ferrules, scales, pins, spacers and transitions for movement or gaps before a small problem becomes structural.

A good kitchen knife should not go in a dishwasher regardless of handle material. Heat, aggressive detergents, impacts against other objects and prolonged hot humidity can damage the handle, assembly and cutting edge.

Material sets possibilities for durability, weight and appearance. What the hand feels comes from the complete combination of material, shape, finish, assembly and care.

Source note: Xinzuo material names and series examples on this page are based on Xinzuo's 2025 product information. PP and ABS are included as general industry comparisons. Natural materials vary, and the individual model specification takes precedence.

Chapter 14

Damascus Steel - Pattern, History and Construction

The word *Damascus* describes several visibly patterned materials that do not share one construction. The safest way to understand a knife is to ignore the pattern for a moment and ask what metal actually forms the cutting edge.

That question separates four families:

Construction	What reaches the cutting edge	What creates the visible pattern
Patterned cladding around a core	A separate named core steel	Welded outer layers exposed by grinding, polishing and etching
Full pattern-welded blade	The alternating welded steels themselves	The arrangement and later exposure of those layers
Crucible-pattern steel	One high-carbon steel mass with an internally developed structure	Bands and particles formed through solidification and controlled forging
Surface-decorated blade	The underlying blade steel	A design applied only to the exterior

These families may look related, but they answer different technical questions.

Pattern is evidence to investigate, not proof of performance.

14.1 Start at the cutting edge

On a clad knife, look for the line where the outer material meets the core. The core continues to the apex and performs most of the cutting work. Its composition, heat treatment, hardness and geometry matter more to edge behaviour than the decorative outer pattern.

Cladding still has practical value. Softer outer material can support a hard core, simplify grinding and manufacturing, change stiffness and weight, and protect part of the blade body. Patterned cladding also gives the knife its visual identity.

On a full pattern-welded knife, alternating steels continue into the edge region. Their compatibility, relative hardness, wear behaviour, layer thickness and orientation can therefore affect the apex directly. The next chapter explains how unequal wear between layers can influence cutting behaviour.

Neither construction is automatically superior. The first uses a specialized core inside a layered body. The second makes the layered material itself the working blade. Both still depend on clean welds, controlled heat treatment and suitable geometry.

14.2 How a welded pattern becomes visible

A smith or industrial producer joins compatible steels into a billet. The billet can then be cut, restacked, twisted or otherwise displaced to redirect the layers. Forging consolidates and shapes the material; grinding cuts through it at different depths.

Polishing first makes the surface geometrically clean. Controlled etching then increases contrast because the steels do not react at the same rate. Nickel-rich layers commonly resist the etchant more strongly and remain brighter, while other layers darken more. The etchant reveals differences already present inside the billet.

The final appearance depends on the entire route:

- steel selection;
- starting layer arrangement;
- manipulation of the billet;
- forging and welding quality;



Figure 14.1: Examples of Xinzuo patterned blades. Use the image to compare visible designs, then use the construction table above to determine what the

- the depth and angle of grinding;
- polishing;
- the chosen etchant and exposure;
- any later surface treatment.

This is why two blades carrying the same layer count can show different line spacing, contrast and movement.

14.3 What layer count can tell us

Numbers such as 33, 67, 73, 101 or 110 describe how many layers the maker counts in the stated construction. They can help explain pattern density and manufacturing design.

They do not reveal:

- which steel forms the edge;
- the composition of the other layers;
- weld integrity;
- heat-treatment quality;
- thickness behind the edge;
- sharpening quality;
- toughness;
- corrosion behaviour;
- whether the count refers to cladding or to a full pattern-welded blade.

More layers can make the visible structure finer or more crowded. They do not create an automatic performance ladder. A lower-layer knife with better heat treatment and geometry can outperform a higher-layer knife.

14.4 Performance follows construction

For a clad blade, begin with the core. A ZDP-189, SRS13, M390, 14Cr14MoVNb, AUS-10 or 10Cr15CoMoV core retains its own metallurgical character even when the surrounding pattern looks similar. The outer material contributes to the body of the blade, but it does not replace the core at the apex.

For a full pattern-welded blade, identify both working steels and how they meet at the edge. Differential wear can preserve microscopic cutting aggression when the layer system and geometry are designed for it. This effect comes from the material that reaches the apex, not from the visible waves higher on the blade.

In either case, the food experiences the complete knife: edge finish, thickness, convexity, blade height, surface texture and movement. A deep etch may change friction slightly. A polished face may move differently through sticky food. Neither guarantees food release.

14.5 Surface decoration is a separate category

A patterned exterior can be produced without building the blade from welded layers. Engraving, coating, controlled coloration and selective surface treatment can create a legitimate decorative result.

The description must match the structure. If the design exists only at the surface, call it a decorated or patterned finish. If welded layers run through the metal, call it pattern-welded. If patterned cladding surrounds a core, name the core and cladding separately.

Surface decoration may fade or change with deep polishing because it does not extend through the blade. A genuine welded pattern remains part of the material, although refinishing can reduce its visible contrast until it is polished and etched again.

14.6 Historical crucible steel is a different route

Historical watered blades associated with the Damascus name were made from high-carbon crucible steel commonly called **wootz**. Their pattern did not come from stacking alternating sheets.

Wootz solidified as an ingot or cake. Composition, cooling and later forging produced bands rich in iron carbide within the steel. Skilled forging preserved and developed that internal structure into the watered appearance visible on finished blades.

The Metropolitan Museum of Art places the origin of wootz production in South India during the middle of the first millennium BCE. Ingots travelled through trade networks into Central Asia, Persia, the Middle East, China and Europe. Damascus became one important centre associated with blades made from this traded material, which helped attach the city's name to watered steel in European usage.

Modern pattern welding and historical wootz can both show flowing lines, but the lines arise differently:

- **pattern welding**: separate steels are joined and mechanically arranged;

- **wootz:** one crucible-steel ingot develops bands through its composition, solidification and forging history.

A 2006 microscopy study reported carbon nanotubes and cementite nanowires in one seventeenth-century Damascus sabre sample. That observation belongs to one blade and one study. It cannot be used to describe every historical watered steel or as a simple explanation of its performance.

14.7 Bintie and the Chinese part of the story

Chinese texts use **Bintie** for prized iron or steel associated with exceptional blades and, in some accounts, a watered appearance. The term appears in Northern Wei records and later sources, but it does not identify one fixed recipe.

Material, finished objects and metallurgical knowledge moved across Eurasia. Some Bintie was described as arriving from Persia or Central Asia; Chinese workshops also had long traditions of crucible steelmaking, co-fusion and controlled processing. Historian William Lox argues that imported and Chinese-made crucible steels may have existed at the same time.

Co-fusion combines higher- and lower-carbon iron materials to obtain a useful final composition. It is another reminder that historical names can describe valued results and trade traditions rather than one standardized modern grade.

Bintie therefore belongs to Chinese metallurgical history without requiring the claim that every example came from one place or used one process. It shows Chinese makers and users participating in a wider exchange of advanced blade materials.

14.8 From the historical question to Xinzuo

Xinzuo does not reproduce an undocumented ancient recipe. Its kitchen knives use specified modern alloys, controlled furnaces, vacuum heat treatment, deep-freeze processing in selected series, industrial grinding and testing.

The historical and modern traditions share a design question rather than an uninterrupted production line: how can composition and processing combine cutting ability, durability and visual character?

Xinzuo answers that question through several structures:

- patterned cladding around cores such as 10Cr15CoMoV, 12Cr18MoV, AUS-10, SRS13, M390, ZDP-189 and 14Cr14MoVNb;

- copper-accented Damascus constructions;
- customized 110-layer full-Damascus blades whose layers continue into the edge.

The Xinzuo Blade Steels guide explains the core materials. Self-Sharpening Full Damascus explains differential wear when alternating materials reach the apex.

14.9 A reliable reading sequence

When examining a patterned knife:

1. Find what forms the apex: a separate core, alternating layers or one crucible steel.
2. Identify the steels and verified heat treatment.
3. Read thickness, grind and edge geometry.
4. Check whether the visible design runs through the metal or sits only on the surface.
5. Interpret layer count and pattern only after the working structure is clear.

This order prevents appearance from standing in for engineering.

Practice principle: identify the working steel first; interpret the pattern last.

Chapter 15

The Self-Sharpening Effect in Full Damascus Blades

Self-sharpening does not mean the knife sharpens itself or replaces worn steel. In selected full-Damascus blades, it describes **differential wear**: alternating steels wear at slightly different rates, helping the edge preserve a microscopic tooth-like structure during use.

Imagine a road made from alternating strips of two materials. If one wears faster, it becomes slightly lower while the harder strip remains more exposed. The surface is no longer perfectly level.

A similar effect can occur at the knife edge. The microscopic “teeth” are too small to see in normal use, but they help the blade grip food and retain slicing bite.

15.1 Why full Damascus matters

The layers must reach the cutting edge.

In a common Damascus-clad knife, patterned layers surround a separate core. The core alone forms the apex, so the cladding cannot create differential wear there even though it may be beautiful and structurally useful.

In full Damascus, the multilayer material forms the blade itself and the alternating steels continue into the edge. If those steels have different wear behaviour, their interaction affects how the apex changes during use.

The term therefore applies to a specific construction, not every knife with a Damascus pattern.

15.2 The role of different steels

The selected Xinzuo full-Damascus blades discussed here pair steels with different hardness and wear characteristics to create this effect.

The more wear-resistant layers recede more slowly; the less wear-resistant layers recede faster. The harder-wearing material remains microscopically more prominent and helps preserve an active slicing bite.

Hardness is only part of the mechanism. Matrix hardness, carbide type and volume, carbide distribution and heat treatment also affect wear resistance. What matters is the actual wear behaviour of the alternating materials after heat treatment, not simply the difference between two HRC values.

The effect also depends on layer thickness and orientation, forge-weld quality, edge geometry and use. The material system and construction create it; the visible pattern alone does not.

15.3 What happens during use

Every cutting edge changes with use. The apex gradually rounds, bends, wears or loses microscopic fragments. Board contact removes material, while food creates friction along the edge.

A uniform-steel edge wears according to one material. In selected full Damascus, the less wear-resistant layers recede faster while the more wear-resistant layers remain slightly more prominent. This preserves a fine saw-like character that helps the edge catch food and begin a slicing cut.

The effect is especially useful during slicing, where the blade moves forward or backward through skins, fibres and other resistant surfaces.

15.4 Sharpness and cutting aggression are different

A highly polished edge can be extremely sharp. Its thin, smooth apex may perform beautifully in a straight push cut.

A toothier edge behaves differently. Microscopic irregularities grip the surface instead of sliding over it, producing strong **cutting aggression**, or **bite**, during slicing.

Differential wear helps preserve this bite. It does not make the apex progressively thinner as the knife is used. *Self-sharpening* means **preserving slicing aggression through differential wear**, not regenerating the edge.

15.5 What self-sharpening does not mean

The knife still needs sharpening. Differential wear cannot repair chips, straighten a badly rolled apex or reverse damage from glass, ceramic, frozen food, hard bone or strong lateral stress.

A suitable full-Damascus construction can keep the edge effectively aggressive for longer between sharpening sessions. It does not make the knife maintenance-free.

15.6 Comparing it with a serrated knife

A bread knife has large, visible teeth that can continue starting a cut even after the points are no longer perfectly sharp. This gives long-lasting slicing aggression.

Full-Damascus differential wear works on a much smaller scale. There are no visible serrations, and the blade still behaves like a fine kitchen knife. The microscopic tooth-like character develops because the materials at the apex wear differently.

A bread knife has teeth ground into its geometry. A full-Damascus blade develops microscopic variation from differential wear.

15.7 Why use still matters

Edges wear differently in different kitchens. Gentle slicing on wood is not the same as hard chopping on plastic. Fibrous vegetables, meat, skins and soft foods also stress the edge differently.

Steel pairing, heat treatment, geometry, board material, food and technique all affect the rate of differential wear. The benefit therefore cannot be reduced to a fixed number of extra weeks or months between sharpenings.

15.8 Sharpening a self-sharpening full Damascus blade

Sharpen the blade when it stops cutting cleanly or begins to require noticeably more force.

Normal stone sharpening restores the apex. Because the layers continue through the blade, sharpening exposes them again.

Follow the existing bevel unless you intend to change the geometry. Use abrasives suited to the steels, remove only enough material to restore the apex, form and remove the burr cleanly, and choose the final grit for the intended cutting style.

A highly polished finish may initially reduce the toothy feel. Microscopic bite develops again as the alternating materials wear at different rates.

15.9 What this means for a Xinzuo knife

Within Xinzuo's range, this explanation applies specifically to selected customized 110-layer full-Damascus blades whose alternating steels reach the cutting edge. It does not apply to a knife simply because it has visible Damascus cladding; in a core-and-cladding blade, the separate core forms the apex.

The feature is straightforward: alternating steels wear at different rates. The less wear-resistant material recedes faster while the more wear-resistant material remains microscopically more prominent, helping the edge keep a natural slicing bite between sharpening sessions.

The key distinction is **differential wear**, not **self-repair**.

In a properly designed full-Damascus blade, alternating materials wear at different rates and help preserve microscopic slicing bite. This can extend useful cutting aggression between sharpening sessions, but normal sharpening remains necessary.

Chapter 16

Routine Knife Care: What to Do Between Sharpenings

Sharpening is only one part of knife maintenance and usually not the most frequent one. Edge life depends more on everyday habits: **what the knife cuts on, how it is washed and dried, how it is stored and whether the edge is used for jobs it was not designed to do.**

Good routine care slows corrosion and mechanical damage and reduces how often sharpening must remove steel.

This chapter covers ordinary maintenance. Full edge restoration appears later in the sharpening chapters.

16.1 After every use: wash, rinse and dry

Wash a quality kitchen knife by hand soon after use. Warm water, mild detergent and a non-abrasive sponge are enough for normal food residue.

Do not leave the knife soaking in a sink. Besides increasing corrosion risk, a sharp blade hidden under cloudy water is dangerous to anyone reaching into the basin.

Rinse salt and acidic foods promptly. Tomato, citrus, vinegar, onions, brine and similar ingredients can accelerate staining or corrosion when left on the blade.

Dry the blade immediately with a clean cloth, moving from the spine toward the edge rather than wrapping fingers around it. Dry the handle and the area around the bolster or ferrule too.

16.2 Why dishwashers are poor for quality knives

A dishwasher combines aggressive detergent, prolonged moisture, heat and the risk of the blade striking cutlery or the basket. Repeated hot, wet cycles can also damage natural wood and some composite or adhesive assemblies.

Hand washing is the safest general rule for quality kitchen knives, even when the steel itself is highly corrosion resistant.

16.3 Stainless steel still needs care

Stainless steel tolerates moisture better than reactive carbon steel, but *stainless* does not mean *corrosion-proof*. Salt, chlorides, acidic residue and long wet storage can still cause staining or pitting.

Treat a small stain as a maintenance signal, not proof of defective steel. Clean and dry the blade, identify whether salt, acid or prolonged moisture caused it, and avoid repeating the exposure.

16.4 Carbon steel, patina and rust

Reactive carbon steel needs faster cleaning and drying. Normal use may create a grey, blue or dark **patina**, a stable surface reaction that develops on reactive steel. A patina is different from destructive orange rust.

Remove active reddish or orange corrosion before it becomes deeper pitting. Start with the least aggressive method suitable for the finish. A mild non-abrasive metal cleaner or very fine rust-removal product is usually safer than coarse scouring on polished or patterned blades.

Deep pitting, damaged finishes or uncertain treatment on a valuable blade are better handled professionally than with aggressive home polishing.

16.5 The cutting board is part of edge maintenance

The edge touches the board hundreds or thousands of times between sharpenings, so board material matters greatly.

Suitable wood and knife-friendly plastic or rubber let the apex enter or deform the surface slightly instead of striking something much harder. Glass, ceramic, stone and metal are poor cutting surfaces for fine edges because they accelerate rolling, flattening and chipping.

Use a board large enough to keep both food and movement stable. Repeatedly striking a plate, pan rim or stone counter can damage an edge much faster than ordinary food.

16.6 Do not scrape the board with the cutting edge

Dragging the cutting edge sideways across the board to gather food loads the apex laterally and gradually rolls or abrades it.

Use the **spine** or **broad face** instead, or use a bench scraper. This simple habit preserves a fine edge surprisingly well.

16.7 Storage: protect the edge from hard objects

Do not leave a knife loose in a drawer where the edge can strike other tools or someone can reach onto it blindly.

Useful storage options include:

- a well-designed knife block;
- a magnetic holder that supports the blade securely without scraping the edge;
- an in-drawer knife tray;
- a properly fitting edge guard or sheath, including the rigid knife covers often called **saya**;
- a protected knife roll for transport.

Dry the knife completely before placing it in a closed guard or sheath. Trapped moisture can create the corrosion environment the guard was meant to prevent.

With a magnetic rack, place and remove the knife under control. Touch the spine or flat to the holder first and rotate the edge away instead of snapping the edge against the surface.

16.8 Natural-wood handles

Do not soak natural wood. Wash it quickly, dry it immediately and avoid standing water and dishwasher heat.

If the handle becomes visibly dry, apply a small amount of **food-safe mineral oil or the treatment recommended by the manufacturer**. Let it absorb, then wipe off the excess before use.

Ordinary cooking oils are not a reliable substitute because some oxidize, become sticky or turn rancid. Finished or stabilized wood may need little or no oil, so the specific handle construction takes priority over a generic routine.

16.9 Composite handles

Pakka wood, Micarta, G10, resin composites and carbon-fibre composites are generally more stable around moisture than untreated wood. They still benefit from hand washing and drying. Moisture stability does not make the whole knife dishwasher-proof, and harsh chemicals can still damage finishes, adhesives or decorative parts.

Inspect the handle occasionally for cracks, looseness or gaps where food and moisture can collect.

16.10 What about oiling the blade?

A stainless kitchen knife that is washed, dried and used regularly usually needs no oil for normal indoor storage.

Reactive carbon steel, long-term storage or humid conditions may justify a very thin protective film. Use only a food-safe product suitable for kitchen blades and wipe off excess before using the knife with food.

16.11 Honing is not sharpening—and rods do different things

The word *honing* can describe several different mechanisms.

A smooth steel can help straighten or recondition a rolled edge on suitable steels whose edge tends to bend rather than chip. Grooved steels, ceramic rods and diamond-coated rods can also remove material, sometimes substantially. They should not all be described as tools that simply “realign” the edge.

Very hard, thin knives that tend to chip rather than roll may gain little from forceful steeling and can be damaged by poor technique. Choose the maintenance tool according to steel, hardness and edge geometry.

When the apex is genuinely rounded, worn or chipped, a rod cannot replace proper sharpening. Your First Successful Hand Sharpening explains how to rebuild the edge.

16.12 Inspect before damage becomes serious

Every few uses, or whenever the knife feels different, inspect it under good light.

Check for:

- bright spots along the apex that indicate dull or rolled sections;
- chips or dents;
- a bent or damaged tip;
- active rust or pitting;
- cracks or looseness in the handle;
- gaps around the bolster, ferrule or scales;
- a wet or contaminated sheath or guard.

Small problems are easier to correct than large repairs. Continuing to use a chipped edge, loose handle or badly rolled apex can worsen damage and create a safety risk.

16.13 Routine care is mostly prevention

A working knife will develop fine scratches, patina, handle wear and other signs of use. Maintenance is not about keeping it visually untouched; it is about preventing normal wear from becoming corrosion, edge damage, structural looseness or unsafe handling.

The everyday routine is short:

1. use the right knife and a knife-friendly board;
2. do not twist, pry or scrape with the cutting edge;
3. wash by hand soon after use;

4. rinse away salt and acids;
5. dry blade and handle completely;
6. store the edge protected and dry;
7. inspect occasionally;
8. sharpen when the edge genuinely needs restoration.

These habits take little time and can save much more work at the next sharpening session.

Chapter 17

Safe Use and Carrying

A good kitchen knife cuts with very little effort. That is why safety must come from habits, not hesitation.

People often say a sharp knife is safer than a dull one. The idea is useful but incomplete.

A sharp knife usually needs less force, enters food more easily and follows the intended line more predictably. A dull knife often encourages harder, less controlled movement. But a sharp knife can still cause serious injury if handled carelessly.

Safety comes from a suitable sharp knife, a stable workspace, controlled technique and clear habits.

Use one short check before every session:

Stable board. Stable food. Clear path. Hidden fingertips.

All four matter. A perfect claw grip cannot rescue a rolling onion, and stable food is not enough if the board slides.

17.1 Create a safe working area before cutting

Many accidents begin before the blade touches food.

Stabilize the cutting board with a damp cloth, non-slip mat or suitable support if needed. Use a board large enough for both the ingredient and the knife. A

board that is too small forces food, blade and hands into the same limited space.

Keep the work area clear. Bowls, packaging, utensils and scraps should not block the cutting path. Keep the floor and counter dry, and use enough light to see the tip, supporting hand and any hard objects inside the food.

A simple routine is:

1. Stabilize the board.
2. Clear the working area.
3. Dry the hands and handle.
4. Stabilize the food.
5. Choose the correct knife.

These few seconds reduce risk and improve cutting quality.

17.2 Stabilize the ingredient

Round ingredients can roll. Create a flat surface on an onion, potato, pumpkin or melon before making the main cuts, then place that surface on the board.

Divide large or irregular food into manageable sections. Forcing a long knife through something that cannot be held securely increases the risk of slipping or twisting.

Avoid frozen food unless the knife is specifically designed for it. A hard frozen surface can make the edge skid, chip or break through suddenly with uncontrolled force.

17.3 Use the right knife for the task

Kitchen knives are not universal cutting tools.

A thin gyuto, santoku or nakiri is excellent for food preparation but not heavy bone. A flexible fillet knife follows contours but should not be twisted against a hard joint. A deba can handle appropriate fish bones but is not a universal cleaver.

Do not use a kitchen knife as:

- a pry bar;
- a screwdriver;
- a can opener;
- a tool for separating frozen objects;

- a lever for opening shells or lids;
- a scraper with the cutting edge dragged sideways across the board.

The steel and geometry are designed for cutting. Sideways force can damage an edge even when straight cutting force is safe.

17.4 Grip and control

Safety does not depend on one universal knife-hand grip. The **handle grip, intermediate or hybrid grip, and pinch grip** described in Weight and Balance can all be controlled when they suit the knife and task.

Whatever grip you use, hold the knife securely without unnecessary tension, keep fingers from sliding toward the edge and keep the wrist in a natural line. If a new grip changes the balance you feel, slow down until the knife becomes predictable again.

The supporting hand should guide the food while keeping fingertips out of the cutting path. In the **claw** grip, fingertips curl inward and the side of the blade moves against the first joints. The claw is a moving guide, not a pose to imitate at professional speed.

How to Cut explains the movements themselves. The safety rule is simpler: use a stable grip, keep the supporting hand behind the cutting line and never trade control for speed.

17.5 Let the edge do the work

A well-sharpened knife should not need excessive force.

If the blade does not enter easily, stop and find the reason. The edge may be dull, the food unstable, the knife unsuitable or the movement wrong for the profile.

Excessive force can make:

- the knife slip;
- the food move suddenly;
- the edge wedge and release unpredictably;
- the user twist the blade while freeing it;
- the supporting hand move into the path.

Use smooth, deliberate movement and avoid lateral stress.

17.6 Use a suitable cutting board

Wood and suitable plastic boards are generally gentle enough for quality knife edges.

Glass, ceramic, stone and metal are too hard for fine edges. Repeated contact can dull or chip the apex quickly. Decorative glass boards may be easy to clean, but they are poor cutting surfaces.

Maintain the board too. Deep grooves can trap residue and affect stability. Clean and dry wooden boards properly instead of soaking them.

17.7 Avoid twisting and sideways scraping

This matters especially with Xinzuo's higher-hardness steels. ZDP-189, SRS13, M390, Blue Steel No. 2 and 14Cr14MoVNB are chosen for demanding edge performance, but a thin hard edge is not designed to pry, twist through a cut, strike bone or cut frozen food. More forgiving steels and heavier geometries offer a larger margin, but they still do not turn a chef's knife into a cleaver or pry tool.

A thin knife is strong in its intended cutting direction and much less tolerant of sideways force.

If a blade becomes stuck in dense food, withdraw it carefully and adjust the cut instead of twisting it free.

Do not scrape food across the board with the cutting edge. Use the spine, blade face or a bench scraper instead.

17.8 Wash safely as well as carefully

Routine Knife Care explains material care. During washing, the safety rule is simple: **keep the blade visible and its edge path controlled.**

Never leave a knife hidden under soapy water or mixed with other objects in the sink. Wash it individually with mild detergent and a soft sponge, keeping the edge away from the hand. Dry it immediately from the spine side without wrapping fingers around the edge.

A dishwasher can also hide a sharp blade among other utensils. Hand washing solves both the safety and material-care problems.

17.9 Passing a knife to another person

Whenever possible, do not pass a knife directly hand to hand.

Place it on a clear surface with the handle accessible and tell the other person it is there. If a direct handover is unavoidable, communicate clearly, move slowly and offer the handle while keeping the blade controlled and pointed away from both people.

Never throw, slide or casually reach a knife toward someone.

17.10 Moving with a knife in the kitchen

For short distances, keep the knife low, close to the body and still.

1. Hold the handle firmly.
2. Keep the arm close to your side.
3. Point the tip toward the floor.
4. Keep the edge facing backward or otherwise away from the body.
5. Do not swing the arm.
6. Warn nearby people that you are passing with a knife.
7. Walk slowly; never run.

Pointing the tip down reduces the chance of directing it toward another person's upper body. Keeping the edge away and the arm still reduces accidental contact.

Do not cover the tip with the free hand or carry the knife upright near the face or chest.

In a professional kitchen, use a clear warning such as “knife behind” or the workplace's established expression.

17.11 Transport beyond the workstation

Do not carry a knife loose through a building, place it unprotected in a bag or leave it in a drawer with the edge exposed.

Use:

- a blade guard;
- a saya;
- a fitted case;
- a knife roll;

- secure packaging designed for transport.

The protection must prevent the knife from sliding out and the edge from cutting through the material.

For transport outside the kitchen, follow local laws and venue rules.

17.12 Safe storage

Store knives with the edge protected and the handle accessible without touching the blade.

Suitable options include:

- a well-designed knife block;
- a stable magnetic strip installed correctly;
- an in-drawer organizer;
- a blade guard or saya;
- a knife roll for professional transport.

A loose knife in a drawer is unsafe and also damages the edge against other utensils.

On a magnetic strip, place the spine against the strip first and rotate the blade into position. To remove it, rotate the edge away before pulling. This avoids striking or scraping the edge against the magnet.

17.13 What to do if a knife falls

Do not try to catch it.

Move your feet away and let the knife fall. Once it stops moving, pick it up by the handle and inspect it for damage.

17.14 Sharpness and safety

Sharpen a dull knife instead of compensating with force.

Signs that the edge needs attention include:

- sliding on tomato skin;
- crushing herbs instead of cutting cleanly;
- needing pressure to begin a cut;
- tearing paper or food;

- moving unpredictably across the ingredient.

Sharpness improves control, but it does not replace safe technique.

17.15 Common unsafe habits

17.15.1 Leaving a knife in the sink

A hidden blade can cut anyone reaching into the water. Keep the knife visible until it is washed, dried and stored.

17.15.2 Cutting toward the body

Direct cuts away from hands and body whenever possible. If specialist in-hand work is appropriate, use a small suitable knife and keep the movement short and controlled.

17.15.3 Holding food in the air with a large knife

Use a cutting board unless the task specifically calls for controlled in-hand work with a suitable small knife. A large blade and unsupported food create unnecessary risk.

17.15.4 Testing sharpness by sliding a finger along the edge

Never slide a finger lengthwise along the cutting edge. Use visual inspection or a controlled cutting test instead.

17.15.5 Scraping the board with the cutting edge

Use the spine, blade face or a bench scraper. Sideways scraping damages the apex and creates unnecessary lateral movement near the guiding hand.

17.15.6 Using a glass, ceramic, stone or metal board

These surfaces rapidly damage fine edges and can make cutting less predictable. Use wood or suitable plastic.

17.15.7 Trying to catch a falling knife

Step back and let it fall. A damaged knife can be repaired or replaced; a hand injury is more serious.

17.16 What to remember

A sharp knife is easier to control because it needs less force, but safety depends on the whole working method. Stabilize the board and food, use the correct knife, keep the supporting hand clear, avoid twisting and carry the knife low and still with the tip down and the edge away.

The safest cook is not the fastest. It is the one whose movements remain predictable.

Practice principle: make every knife movement easy for another person to anticipate.

Chapter 18

Your First Successful Hand Sharpening

The aim of a first sharpening session is simple: make the whole edge cut better from heel to tip.

You do not need a perfect angle, an expensive collection of stones or fast movements. You need one stable stone, one repeatable motion and clear signs that tell you what to do next.

This chapter deliberately teaches a forgiving method. Every normal pass covers the complete edge, from tip to heel. An experienced sharpener may work only on the area that needs correction, but a beginner can easily create sharp and dull zones that way. Removing a tiny amount of extra steel is a reasonable price for a first edge that works evenly.

Use this method on an ordinary double-bevel kitchen knife: a chef's knife, santoku, utility knife, nakiri or paring knife sharpened on both sides. Practise on a sound, inexpensive knife before using your most valuable one.

Do not use this method on:

- a serrated knife;
- a traditional single-bevel yanagiba, deba or usuba;
- a bent or twisted blade;
- a blade with a crack, a deep chip or a badly broken tip;
- a knife whose handle or bolster prevents the heel from reaching the stone.

These cases need a different technique or a repair before normal sharpening.

18.1 What you are actually sharpening

Three parts are easy to confuse:

- The **primary bevel** is the broad surface that makes the blade thinner toward the edge.
- The **microbevel** is the very narrow terminal facet immediately above the cutting edge.
- The **apex** is the line where the two microbevels meet. This is the part that cuts.

In this beginner method, sharpen only the **microbevel**. Do not lay the broad primary bevel on the stone. Grinding the primary bevel changes the thickness and geometry of the blade. That operation is called thinning and belongs to the next chapter.

A dull apex has become rounded, bent or worn. The stone removes steel from the two microbevels until they meet again in a clean line.

When the stone reaches the apex from one side, it often pushes a tiny lip of steel toward the other side. This lip is the **burr**. A light burr along the entire edge tells a beginner that the stone has reached the apex everywhere.

The burr is a progress signal, not the finished edge. It must be removed before the knife is ready.

18.2 What you need

Keep the first setup small:

- one medium water stone, ideally around 800-1200 grit;
- a stone holder, rubber base or damp folded cloth;
- water and a towel;
- a dark permanent marker;
- a pencil for checking stone flatness;
- ordinary printer paper for testing.

A 1000-grit stone is the simplest starting point for a normally dull kitchen knife. It removes steel quickly enough to reach the apex while remaining controllable.

A very coarse stone removes mistakes as quickly as it removes damage. Keep stones below about 500 grit for chips, profile repair and later training. A very fine stone can polish the microbevel without reaching a rounded apex efficiently. It is optional during the first sessions.

If the knife remains unchanged after many correct passes on a medium stone, stop before pressing harder. The knife may need a more effective abrasive, a coarser stone or a repair diagnosis. The next chapter explains those choices.

18.3 Inspect the knife before sharpening

Clean and dry the blade. Look straight along the cutting line and then tilt the apex slowly under a strong light.

Check for:

- bright points or short bright lines along the apex;
- chips or dents;
- a rounded or damaged tip;
- a bend, twist or local hollow in the cutting line;
- corrosion close to the edge;
- a microbevel that is much wider on one side;
- a heel that cannot lie on the stone because the handle or bolster blocks it.

Small reflections normally show dull or rolled areas. Deep chips, cracks, obvious bends and severe profile damage are stop signs for the beginner method.

Make a slow paper cut with the heel, middle and tip separately. Remember or write down where the knife catches. This is your starting map. It lets you compare the same areas after sharpening instead of relying on memory.

18.4 Prepare the stone and your workspace

Read the instructions supplied with the stone. Some water stones need soaking; others need only water on the surface. Soak or wet the stone exactly as its maker directs. Do not assume that every water stone can remain immersed. When an absorbent stone does require soaking, the slowing of air bubbles is a useful sign that its pores have filled with water; the maker's instructions still take priority.

Place the stone lengthwise on a stable support. Push gently on each corner. If it rocks or slides, fix the support before touching the knife to it.

The surface must be reasonably flat. Draw a loose pencil grid across the stone. Rub it against the recommended flattening plate until the lines disappear together. If the centre retains pencil while the ends become clean, the stone is hollow and needs more flattening.

Keep the stone wet during sharpening. Add water when the blade starts to drag, the sound becomes dry or the surface begins to feel sticky. The grey water is normal. It contains loose abrasive and steel removed from the knife; it is not the burr.



Figure 18.1: A stable stone is more important than a fast hand. Secure it before trying to control the angle.

Stand or sit so your shoulders remain relaxed and your elbows can move freely. The stone should be high enough that you do not bend deeply over it, but low enough that your shoulders do not rise. Leave enough empty space to complete the whole pass without hitting the sink, wall or other objects.

18.5 Find the microbevel with a marker

Look for the very narrow shiny strip immediately above the cutting edge. This is the microbevel. On some knives it is easy to see; on others it may be less than a millimetre wide.

Colour only the last one or two millimetres nearest the edge. Do not colour

the broad primary bevel as if it were the surface to sharpen.

Rest the blade lightly on the stone. Raise the spine until the microbevel seems to meet the surface, then make one nearly weightless complete-edge pass. Inspect the ink.

- Ink removed above the microbevel but left at the apex means the spine was too low. The stone did not reach the cutting edge.
- Ink removed only from the extreme apex means the spine was higher than the existing microbevel angle.
- Ink removed across the microbevel and up to the apex means the contact is correct enough to begin.

The exact appearance does not need to be perfect. The important point is that the stone reaches the apex without scratching a wide band across the primary bevel.

Around 15 degrees per side is common for many kitchen knives, but it is not a universal specification. A useful visual approximation is to lift the spine by about one quarter of the blade height. For a 50 mm tall chef's knife, this is about 13 mm. A fixed stack of two coins cannot represent the same angle on knives of different heights and should not be used as a universal gauge.

Use the height estimate only to understand the scale of the movement. The marker tells you more because it reads the microbevel that is already on the knife.

18.6 Learn one complete-edge pass

Hold the handle with your dominant hand. Place two or three fingertips from the other hand on the flat of the blade, safely behind the cutting edge. For the beginner method, keep them near the centre of the blade.

The fingertips rest on the blade and move together with it. Do not slide them from tip to heel while the knife is moving. Local finger pressure will be introduced only after your full-edge movement is reliable.

Start with the tip on the far part of the stone. Because the edge curves upward near the tip, begin with the handle already raised slightly. Draw the knife diagonally across the stone so the contact travels through this sequence:

tip - curved front - middle - heel.

As contact moves from the tip through the curved front, lower the handle gradually. Keep its height almost unchanged as the straighter middle and heel

cross the stone. At the end, remove pressure, lift the knife, return to the starting position and repeat.

During the working movement, let the cutting edge travel ahead of the spine, as if trying to slice an impossibly thin layer from the stone. This is called an edge-leading pass. Keep the angle stable and do not let the edge dig into the surface.

Do not grind during the return movement in the beginner method. One deliberate edge-leading movement followed by one unloaded return is easier to control than rapid movement in both directions.

Every normal working pass must include the entire edge. Use the same path each time. Move slowly enough to notice sound, resistance and contact. Speed hides mistakes and adds nothing to the result.

18.6.1 How much pressure to use

Use three simple pressure levels:

1. **Working pressure:** enough to hear and feel the stone cutting, but not enough to bend the blade, move the stone, whiten the fingertips or tighten the shoulders.
2. **Refining pressure:** about half the working pressure once the burr is continuous.
3. **Finishing pressure:** little more than the weight of the knife and the hands guiding it.

If the knife sticks, shakes or changes angle, pressure is already too high. Add water, relax and slow down before trying again.

18.7 The complete beginner method

18.7.1 Step 1: make the starting map

Inspect the clean edge under light and test heel, middle and tip on printer paper. Do not sharpen a cracked, bent, deeply chipped or unusual blade.

18.7.2 Step 2: mark the first microbevel

Colour only the narrow microbevel. Make one light complete-edge pass and adjust the spine until the ink disappears from the microbevel up to the apex from tip to heel.

Do not continue if the stone is scratching the broad primary bevel. Recolour the last one or two millimetres and repeat the contact check.

18.7.3 Step 3: sharpen the first side

Make five slow complete-edge passes. Start with the handle slightly raised at the tip, lower it through the curve and finish at the heel. Keep the same microbevel angle and use controlled working pressure.

Rinse the edge and check for a burr on the side opposite the one you sharpened. Check at the heel, middle, curved front and tip.

If the burr is missing anywhere, make another two or three complete-edge passes. Continue to cover the whole edge, but pay closer attention as the missing area crosses the stone. Inspect again after each small group.

Stop as soon as the first light burr is continuous. Do not make it larger for reassurance.

18.7.4 Step 4: sharpen the second side

Turn the knife over while keeping the handle in your dominant hand. Place the other fingertips safely on the flat and repeat the same complete path from tip to heel.

Work first in a group of five and then in groups of two or three. Stop when the burr has moved to the first side along the complete edge.

The second side may need a different number of passes. Stroke counts create rhythm; the burr tells you whether the apex has been reached.

18.7.5 Step 5: reduce and remove the burr

Keep the complete-edge movement, but reduce both pressure and repetition:

1. three light passes on the first side;
2. three light passes on the second;
3. two lighter passes on the first;
4. two lighter passes on the second;
5. six to ten nearly weightless passes, changing side after every pass.

Rinse and inspect. If a directional burr remains, do not return to heavy pressure. Continue with one nearly weightless pass per side and check again.

This countdown is a starting routine, not proof that the burr has disappeared. The checks below decide when the knife is finished.

18.7.6 Step 6: use a finer stone only after the medium stone succeeds

The first goal is a stable edge from the 1000-grit stone. Do not move to a finer stone to hide an incomplete microbevel.

If the knife already passes the light and paper checks and you have a stone around 3000 grit, clean the knife, hands and holder. Make only alternating complete-edge passes with finishing pressure. Do not raise another noticeable burr.

If you are unsure what the fine stone changed, stop on the medium stone. A clean, well-deburred 1000-grit microbevel is an excellent everyday kitchen edge.

18.7.7 Step 7: clean before testing

Wash and dry the knife. Remove abrasive residue from the handle, blade and work surface. Carbon steel should be dried immediately.

Rinse the stone and let it dry slowly in a ventilated place as its maker recommends. Do not use an oven, radiator, hair dryer or strong direct sun. Do not close a damp stone inside a box.

Test the knife only after it is clean.

18.8 Understand and check the burr safely

The burr is a microscopic lip still attached to the apex. It bends toward the side opposite the one being sharpened. The burred side may feel slightly rough or hooked when approached from the blade face.

Never slide a finger along the cutting edge.

To check with a fingertip, hold the knife still and move the fingertip very lightly from the blade face toward the edge, perpendicular to it. Stop as soon as you reach the apex. Repeat at the heel, middle, curved front and tip.

If you do not want to place a fingertip near the apex, use a small piece of cotton, tissue or thin paper. A burr may catch loose fibres more readily from one side. Good light or magnification can also help.

Do not confuse the burr with:

- grey slurry or steel residue stuck to the blade;
- rough scratches that feel the same from both directions;
- a rolled section that was already present before sharpening;

- a burr found at only one point, which proves only that one point reached the apex.

A large burr is not better. It can fold from side to side, cut paper briefly and then collapse in the first real use. Treat the burr as a receipt, not a trophy: the smallest continuous one is enough.

18.9 Three checks after sharpening

No single test proves that the whole knife is ready. Use all three.

18.9.1 Check 1: reflected light

Hold the clean, dry apex under a strong lamp and tilt it slowly. A clean apex appears mostly dark. A bright point or short bright line often marks a flat, rolled or chipped area.

One tiny reflection does not justify grinding the whole knife again. Note its position and compare it with the paper test.

18.9.2 Check 2: paper in three sections

Use ordinary dry printer paper. Hold it so your supporting hand remains well away from the cutting path.

Test three overlapping sections separately:

1. heel to middle;
2. middle to curved front;
3. curved front to tip.

Cut slowly instead of slashing. Notice how much force starts the cut, whether resistance changes and whether the paper is sliced cleanly or torn. The three parts should behave similarly. A catch in the same position twice identifies an area that needs attention.

Paper alone is not enough. A wire burr can cut it once and then fold.

18.9.3 Check 3: food and stability

A tomato checks whether the apex starts on smooth skin without sliding. An onion checks entry and movement through layers. A carrot also reveals thickness behind the microbevel, but that subject belongs to the next chapter.

Use the knife normally. Do not force it into food to prove that it is sharp.

After about twenty normal cuts through onion or carrot, clean the knife and repeat the three-part paper test. If the result becomes much worse immediately, a wire burr probably remained. Return only to the nearly weightless alternating passes, clean the knife and test again.

18.10 Workshop examples

18.10.1 The middle cuts, but the tip catches paper

The user held a useful angle on the straight part but did not begin with the handle raised enough to follow the tip.

Recolour the tip microbevel. Make three complete-edge passes per side. Start with the handle slightly raised, then lower it gradually through the curve. Test the tip again.

The beginner still covers the whole edge. Only the attention given to the tip changes.

18.10.2 The knife cuts paper once, then feels dull

A flexible wire burr probably remained. More heavy sharpening would make it larger.

Return to the medium stone and use six to ten nearly weightless alternating complete-edge passes. Clean the knife, make several normal food cuts and test again.

18.10.3 The marker disappears above the microbevel but remains at the edge

The angle is too low. The stone is polishing above the microbevel without reaching the apex.

Raise the spine slightly, recolour the last one or two millimetres and repeat one light contact pass.

18.10.4 The marker disappears only from the very edge

The spine is higher than the existing microbevel angle. This can create a steeper new microbevel instead of following the one already present.

Lower the spine slightly and repeat the marker check. Stop adjusting when the stone reaches the complete microbevel and apex.

18.10.5 The burr appears at the heel and middle but not at the tip

The tip has not reached the apex. Continue with two or three complete passes at a time. Begin with the handle slightly raised and watch the marker at the curved front.

Do not grind only the middle because its burr feels reassuring.

18.10.6 One side forms a burr quickly and the other takes longer

This is common on a previously uneven microbevel. Do not force equal stroke counts. Complete the burr on each side and then use the same light alternating finish.

18.10.7 The blade sticks and the movement becomes jerky

The stone may be dry or clogged, or the pressure may be too high. Add water, rinse the surface if needed and reduce pressure. A slow smooth pass sharpens better than a forceful pass that changes angle.

18.10.8 The knife cuts paper but wedges in a carrot

The microbevel may be sharp while the blade behind it is too thick. Repeated sharpening of the microbevel will not solve that resistance. This is a geometry problem and introduces thinning in the next chapter.

18.11 Mistakes that matter most

18.11.1 Sharpening the primary bevel instead of the microbevel

The primary bevel is the broad surface above the narrow terminal facet. Grinding it changes blade geometry and can damage a polished, hammered or Damascus finish. The beginner method touches only the microbevel.

18.11.2 Changing the angle during every pass

An approximate angle repeated well works better than a perfect angle touched only occasionally. Slow down and repeat the marker check.

18.11.3 Pressing harder when nothing happens

First check microbevel contact, water and stone condition. Pressure cannot compensate for missing the apex.

18.11.4 Stopping the pass in the middle

The complete-edge method works because every normal pass includes tip, middle and heel. Remove pressure, reset and repeat the full path.

18.11.5 Chasing a large burr

A large burr feels obvious but is harder to remove. Stop when the first light burr is continuous.

18.11.6 Testing only the middle

The middle is usually easiest to sharpen. Test the heel, curved front and tip separately as well.

18.11.7 Starting on a damaged or expensive knife

Large chips, unusual geometry and valuable finishes create problems that a first session does not need. Learn the movement on a simple knife first.

18.12 Base-level exercises

These exercises turn a lucky result into a repeatable skill.

18.12.1 Exercise 1: stable setup

Prepare the stone, then push gently on each corner and make ten unloaded practice movements. Pass when the stone stays still, the shoulders remain relaxed and the movement finishes without hitting an obstacle.

18.12.2 Exercise 2: the marker line

Colour only the microbevel. Make three very light complete-edge passes per side. Pass when ink is removed from the microbevel up to the apex at the heel, middle and tip without creating a wide scratched strip across the primary bevel.

If the result varies, recolour and repeat more slowly. Do not continue to burr formation until contact is understandable.

18.12.3 Exercise 3: the burr map

Sharpen one side in small groups and check heel, middle, curved front and tip after every group. Pass when you can identify the first light continuous burr and stop without making it much larger.

18.12.4 Exercise 4: pressure reduction

Say or note the three stages while working: working, refining and finishing. Pass when the final alternating movements are clearly lighter than the first passes and the knife no longer carries a directional burr.

18.12.5 Exercise 5: the three-part paper test

Test heel-middle, middle-front and front-tip. Pass when all three sections start and travel with similar resistance and none repeatedly tears the paper.

18.12.6 Exercise 6: edge stability

Make about twenty normal cuts in onion or carrot, clean the knife and repeat the paper test. Pass when the edge behaves substantially as it did before the food test.

18.12.7 Exercise 7: repeat the result

Sharpen a second ordinary knife, or repeat the process after the first knife has genuinely become dull again. Pass when the result is similar without having to rediscover the movement from the beginning.

18.13 Your base-level test

Move to the next chapter only after you can answer yes to every statement in two different sharpening sessions:

- I can identify a suitable double-bevel practice knife and recognise the cases that require me to stop.
- I can prepare a flat, wet and stable stone without it moving.
- I can distinguish the primary bevel, microbevel and apex.

- I can use marker to place only the microbevel on the stone from tip to heel.
- Every normal working pass covers the complete edge.
- I begin with the handle slightly raised at the tip and lower it through the curve.
- I can form the smallest useful continuous burr and stop.
- I can reduce pressure and remove the directional burr with alternating passes.
- Heel, middle and tip pass the paper test with similar resistance.
- The result remains stable after normal food cutting.

Passing once can be luck. Passing in two sessions shows that the method is repeatable.

At this point you have learned the essential skill: controlled contact on the microbevel along the complete edge. The next chapter teaches when and how to depart from this safe beginner method.

Chapter 19

Improving Your Sharpening Skills

The first chapter gives one dependable answer to most beginner problems: follow the existing microbevel, cover the complete edge, form a light burr on both sides, then finish with very light alternating passes.

Keep that method whenever it gives a good result. Improvement does not mean replacing a simple method with a complicated one. It means noticing more, correcting less and understanding when the simple method needs help.

Begin this chapter only after you can sharpen an ordinary double-bevel knife evenly from heel to tip and repeat the result in two sessions. If microbevel contact, burr detection, the three-part paper test or the stability test is still uncertain, more advanced choices will hide the basic problem rather than solve it.

19.1 The three skills behind better sharpening

An improving sharpener learns to control three things:

1. **Contact:** exactly where the stone touches the knife.
2. **Pressure:** how strongly it cuts and how pressure changes during the session.
3. **Purpose:** whether the work is restoring the apex, correcting the blade behind it or changing the final cutting feel.

These skills reduce unnecessary grinding. They also explain why a knife can have a keen edge and still perform poorly in food.

19.2 Read the evidence after every stage

The stone leaves visible and tactile evidence:

- marker removal shows contact on the microbevel;
- fresh scratches show where steel was removed;
- reflected light reveals remaining flat or damaged areas;
- the burr confirms that the abrasive reached the apex;
- resistance in paper maps uneven sections;
- resistance in food reveals both the edge and the blade geometry behind it.

Do not collect all this information at the end. Stop, rinse and inspect during the work. Early inspection turns a correction into a few controlled passes. Late inspection often discovers that too much steel has already been removed.

19.2.1 Practice drill: predict before looking

Colour only the microbevel of a practice knife. Make one light pass while deliberately holding the spine slightly too low. Before inspecting it, predict where the ink should remain. Then repeat with the spine slightly too high.

This safe marker exercise teaches the connection between hand position and contact. Return to the correct angle and erase the intended strip evenly before sharpening further.

Pass the drill when your prediction matches the mark three times in a row.

19.3 Build three pressure levels

Pressure has different jobs during one sharpening session.

19.3.1 Working pressure

Use controlled pressure to remove steel and reach the apex. The stone should cut steadily without stopping the blade, moving the holder or making the wrist collapse.

19.3.2 Refining pressure

Once the burr is continuous, use about half the previous pressure. The goal is now to replace coarse scratches and reduce the burr, not to keep bending it.

19.3.3 Finishing pressure

The final alternating passes use little more than the weight of the knife and guiding fingers. They clean and centre the apex.

A beginner often applies one pressure from start to finish. The result may feel sharp while still carrying a wire burr. The clearest sign of progress is not faster movement; it is a visible reduction in pressure as the edge nears completion.

19.3.4 Practice drill: make the burr smaller

Sharpen two similar inexpensive knives on the same medium stone. On the first, stop at the earliest continuous burr. On the second, continue for twenty unnecessary firm passes.

Debur both with the same light sequence. Compare how long the process takes and whether the second burr keeps folding from side to side. The extra work usually makes the finish harder rather than better.

The lesson is practical: the smallest burr that confirms complete contact is enough.

19.4 Introduce section work carefully

The complete-edge pass remains the safest default. Section work becomes useful when one area needs more attention, the blade is longer than the stone, or a local defect cannot be corrected efficiently with full passes.

Divide the edge into overlapping zones: heel, middle and front. Work one zone with short controlled strokes while keeping pressure directly above it. Overlap the neighbouring zone so no abrupt step develops. Finish with several light complete-edge passes to blend the whole edge.

Section work saves steel only when the sharpener can preserve the profile. Repeated strokes in one exact spot can create a hollow in the cutting line. Mark the microbevel, inspect often and stop local work as soon as the missing contact or burr appears.

19.4.1 Workshop example: one dull centimetre near the heel

The complete-edge beginner method would sharpen the whole knife until the heel improves. That is safe and may be the correct choice for a new sharpener.

An improving sharpener can make three to five short strokes over the heel, with the supporting fingers above it, then return to light full-edge passes. The correction ends when the burr and paper response match the rest of the blade.

The advanced method is not automatically better. It is better only when the user can keep the correction local without changing the profile.

19.4.2 Section-work test

Colour the full microbevel and leave one small area with extra marker. Work that area briefly, blend with full passes and inspect the edge line from the side.

Pass when the target mark is removed, the neighbouring microbevel remains even and the paper test has no catch at either transition.

19.5 Choose stones by the work

Grit numbers describe abrasive size, but they do not tell the whole story. Stone hardness, binder, abrasive type, cutting speed and slurry change how a stone behaves.

Use a simple three-stage decision:

- **Repair:** a coarse stone, usually below about 500 grit, for chips, profile correction or substantial thinning.
- **Sharpen:** a medium stone around 700-1500 grit for most dull edges.
- **Refine:** a finer stone around 2000-6000 grit when the task benefits from a smoother finish.

The ranges overlap between brands. Begin with the finest stone that can complete the required work in a reasonable time. A coarse stone saves time on damage but removes steel quickly. A fine stone gives control but cannot efficiently rebuild a badly rounded edge.

The abrasive must also cut the steel. Conventional water stones suit many common Xinzuo steels. Highly wear-resistant steels such as ZDP-189, SRS13, M390 and 14Cr14MoVNB may respond more efficiently to suitable ceramic or diamond abrasive. If the stone barely changes the microbevel despite correct contact, improve the abrasive before increasing force.

19.5.1 Hard and soft stones

A hard stone stays flat longer and shows contact clearly. This makes it useful for diagnosis and precise bevels. A softer stone releases abrasive readily and can feel smooth, but it dishes faster and can round a surface under heavy pressure.

Neither is a quality rank. Choose the behaviour that helps the stage you are performing.

19.5.2 Water, slurry and a readable surface

The grey mixture on a water stone contains removed steel and loose abrasive. It is called slurry or swarf depending on what it contains. It is not the burr.

Rinse the surface when you need to see exact contact, when the stone feels sticky or when metal begins to clog it. Keep a controlled slurry when the stone cuts and finishes better with it. A beginner may simply keep the surface wet and clean; deliberate slurry management belongs at this second level because the user can now notice the difference.

Clean the knife, hands and holder before moving to a fine stone. One coarse particle can leave a deep scratch across a polished microbevel or blade finish.

19.6 Keep the stone flat

The blade copies the surface beneath it. A hollow stone changes the angle during a stroke and makes accurate work harder.

Draw a pencil grid across the stone. Flatten until every line disappears. Check more often when using a soft stone, working a wide bevel or repeatedly sharpening the centre. Light, frequent flattening is faster than repairing a deeply dished stone.

19.6.1 Flatness test

Check the stone with pencil before a session, after sharpening two knives and again after flattening. Pass when you can recognise the developing wear pattern and correct it before the hollow becomes obvious under a straightedge.

19.7 Distinguish apex work from geometry work

Routine sharpening restores the narrow microbevel. It brings the two sides together at the apex.

As sharpening repeats, that microbevel moves upward into thicker steel and becomes wider. The knife may cut paper well but wedge in a carrot or onion. The apex is sharp; the blade behind it is too thick.

Thinning removes steel behind the edge to restore easier passage through food. It is a separate operation from sharpening and removes much more material.

19.7.1 A practical geometry comparison

Imagine two pencils sharpened to equally fine points. One tapers gradually; the other becomes thick immediately behind the point. Both points can pierce paper, but the gradual pencil passes through a narrow opening more easily.

A kitchen knife behaves in the same way. Paper mostly tests the point of the system. Firm food also tests the shape behind it.

19.7.2 Recognise when thinning may help

Possible signs include:

- the knife passes paper but splits or wedges in firm vegetables;
- the microbevel has become much wider after repeated sharpening;
- cutting resistance has increased even after a clean deburring;
- more force is needed once the apex has already entered the food.

Before thinning, inspect the blade finish. Damascus pattern, mirror polish, coatings, hammering and black forge scale can be scratched or removed. Protect valuable finishes and use professional service when the original grind is uncertain.

For a simple practice knife, colour the area immediately behind the microbevel. Lay the blade at a lower angle than the sharpening angle and abrade that shoulder gradually. Keep the cutting profile unchanged. Then recreate the narrow microbevel and finish normally.

Do not try to make the visible microbevel equally wide at heel and tip. Blade thickness usually changes toward the tip. Equal appearance can require unequal geometry.

19.7.3 Geometry test

Compare the knife in printer paper and a firm carrot before and after normal sharpening. If paper improves but carrot resistance remains high, identify geometry as the likely next subject rather than moving automatically to a finer stone.

This test is passed when you can explain whether the problem begins at the apex or after the blade has entered the food. Actual thinning should wait until that distinction is clear.

19.8 Choose the finish for the food

Higher grit is not automatically sharper in every useful sense. A medium finish leaves microscopic texture that can start easily in tomato skin, meat and fibrous food. A finer finish can feel smoother in push cuts and delicate slicing.

Use the same knife and compare two finishes over time. Finish it first on the medium stone, deburr carefully and test tomato, onion and paper. At the next genuine sharpening, add the finer stone and repeat the same tests.

Keep the finish that suits the knife's work. Prestige, mirror appearance and grit number matter less than controlled cutting.

19.9 Improve deburring before adding polish

A wire edge can imitate excellent sharpness. It bends or breaks during the first cuts, so the knife seems to become dull immediately.

Prevent it by stopping at a small burr and reducing pressure early. Alternate sides more frequently near the end. Edge-leading strokes, where the edge moves into the abrasive, can cut weakened burr steel away efficiently. Edge-trailing strokes, where the spine leads, feel gentle but can draw out a new wire if pressure is high.

A clean leather or fabric strop is optional. Move spine first, use very few passes and press lightly. A soft strop under heavy force rounds the apex. The stone must do the main deburring; the strop is a final check, not a rescue tool.

19.9.1 Stability test

Cut paper, make twenty normal cuts through cardboard-free food such as onion or carrot, clean the knife and test paper again. Pass when the edge feels

substantially the same and no section develops a sudden catch.

The test is not about long-term edge retention. It checks whether the first result depended on a weak burr.

19.10 Approach asymmetry with evidence

Some double-bevel knives have one side sharpened more than the other. Previous owners may also have created accidental asymmetry.

Do not equalise the two sides by counting identical strokes. Use marker, microbevel width, steering in food and burr movement. Preserve a deliberate asymmetry when the knife cuts straight and the maker specifies it. Correct an accidental one gradually rather than trying to rebuild it in a single session.

If the knife always steers to one side in a tall food such as potato, first confirm that the cut and grip are straight. Then compare microbevel angles, thickness and remaining burr. Steering is evidence, not a command to grind one side heavily.

19.11 Traditional single-bevel knives

A yanagiba, deba or usuba needs a different method. Its broad front bevel works with a hollow reverse called the **urasuki**. Narrow flat lands around that hollow support the edge; light work on them is called **uraoshi**.

Do not apply the beginner's double-bevel method to these knives. The reverse is not sharpened at a normal raised angle. It is placed almost flat on a truly flat stone and touched with very light pressure. Raising it can create a secondary bevel and damage the geometry.

The broad front bevel may also change contact from heel to tip because blade height, thickness and twist vary. A hard medium-fine stone and marker can reveal high and low areas before coarse work begins.

For an improving sharpener, the safe first lesson is diagnosis:

1. flatten the stone carefully;
2. place the reverse flat and make a few nearly weightless passes;
3. confirm that contact stays on the narrow perimeter lands rather than inside the hollow;
4. place the broad front bevel on the stone and inspect its contact pattern;
5. stop if the blade rocks, appears bent or would need extensive coarse correction.

Routine refinement of a sound single bevel can be learned with model-specific instruction. Correcting the reverse hollow, wide lands, serious twist or a damaged ridge line belongs to an experienced specialist.

19.11.1 Hamaguri in plain language

Hamaguri is a controlled convex transition on the front bevel. It can add support near an impact-prone edge, as on some deba sharpening strategies. It is not one compulsory shape for every Japanese knife.

Soft stones, thick slurry and rocking pressure can create accidental convexity. Learn flat contact first. A deliberate small microbevel is easier to control than vaguely rounding the whole surface.

19.12 A five-session improvement plan

19.12.1 Session 1: contact

Use a practice knife and marker. Predict the ink pattern, make one light pass and inspect. Repeat until the hand can reproduce the same narrow contact strip.

19.12.2 Session 2: pressure

Sharpen normally, but name each stage aloud or in notes: working, refining, finishing. The final passes must be visibly lighter than the first.

19.12.3 Session 3: burr control

Check after every group of five passes. Stop at the first continuous burr and record how little work was actually needed. Test stability after deburring.

19.12.4 Session 4: local correction

Use marker to identify one incomplete section. Apply a few overlapping local strokes, then blend with complete-edge passes. Confirm the edge profile and paper response.

19.12.5 Session 5: task finish

Compare a medium and a fine finish on the same type of food. Describe bite, smoothness, resistance and control without using the grit number as the conclusion.

19.13 The second-level test

You have moved beyond beginner technique when you can demonstrate all of the following:

- predict and confirm contact with marker and scratches;
- use clearly different working, refining and finishing pressures;
- stop at a small continuous burr and produce a stable edge;
- correct one section without creating a hollow or transition catch;
- keep the stone flat and free from coarse contamination;
- distinguish a dull apex from excessive thickness behind the edge;
- choose a medium or fine finish by food performance;
- recognise a single-bevel knife and avoid damaging its reverse geometry.

Passing this test does not mean every repair should be attempted. It means you can read what the knife and stone are showing, choose a proportionate action and stop before uncertainty becomes damage.

19.14 What expert practice adds

Expert sharpening uses the same evidence with greater precision. The sharpener may change stroke direction to read scratches, select a hard stone to diagnose contact, use soft-stone slurry to blend a bevel, minimise burr formation or vary support along a twisted single bevel.

Those techniques are valuable because the foundation is stable. Without angle control, pressure control and reliable testing, extra choices only create extra ways to be inconsistent.

Keep the complete-edge method as your safe home position. Depart from it for a clear reason, make the smallest correction that evidence supports, then return to full light passes and test the whole knife.

Chapter 20

Bibliography

All bibliographic, technical and historical sources cited throughout the book are collected here so that the individual chapters remain focused on the explanation itself. Entries are grouped by the chapter in which they are used.

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- Knife Steel Nerds, Carbide Types in Knife Steels
- Knife Steel Nerds, Which Steel Has the Best Edge Retention? Part 2
- Knife Steel Nerds, Which Knife Steels Have the Best Corrosion Resistance?

20.2 Alloying Elements

- worldstainless, Stainless Steel Terminology
- British Stainless Steel Association, When is stainless steel passive or active?
- Knife Steel Nerds, Which Knife Steels Have the Best Corrosion Resistance?
- Knife Steel Nerds, Carbon vs Stainless Steel in Knives

- William Lox, “Bintie: The Wootz Steel in Ancient China,” *Indian Journal of History of Science* 44.3 (2009)

20.3 Conventional and Powder-Metallurgy Steels

- The Hong Kong Polytechnic University, *Professional Guide on Steel*, section 2.2.2: Iron-carbon phase diagram
- Aalto University, Fe-C phase diagram
- Carpenter Technology, Powder Metal Alloy Bridges the Gap
- Carpenter Technology, Micro-Melt 20-4
- Uddeholm, Vanadis 8 SuperClean - The Powder Metallurgy Process

20.4 Heat Treatment and Carbides

- Knife Steel Nerds, Carbide Types in Knife Steels
- Knife Steel Nerds, Cryogenic Processing of Steel Part 1 - Maximizing Hardness
- Knife Steel Nerds, Cryogenic Processing of Steel Part 3 - Wear Resistance and Edge Retention
- Knife Steel Nerds, Dry Ice vs Liquid Nitrogen vs Freezer - How Cold Do You Need for Cryo?
- Knife Steel Nerds, M390 vs 20CV vs 204P
- Xinzuo 2025 product catalogue, individual series specifications for vacuum heat treatment and deep-freeze processing.

20.5 Xinzuo Blade Steels

- Xinzuo 2025 product catalogue and individual current series specifications
- Knife Steel Nerds, Carbide Types in Knife Steels
- Knife Steel Nerds, Cryogenic Processing of Steel Part 1
- Knife Steel Nerds, M390 vs 20CV vs 204P
- William Lox, “Bintie: The Wootz Steel in Ancient China,” *Indian Journal of History of Science* 44.3 (2009)

20.6 Xinzuo Handle Materials - Feel, Care and Choice

- Museum of Fine Arts Boston CAMEO, Micarta

20.7 Damascus Steel - Pattern, History and Construction

- The Metropolitan Museum of Art, Ten Wootz Steel Ingots with Bag
- The Metropolitan Museum of Art, Persian Saber with a Watered-Steel Blade
- William Lox, “Bintie: The Wootz Steel in Ancient China,” *Indian Journal of History of Science* 44.3 (2009)
- Donald B. Wagner, *Iron and Steel in Ancient China* (Brill, 1993), and *Science and Civilisation in China*, Vol. 5, Part 11: *Ferrous Metallurgy* (Cambridge University Press, 2008)
- Marianne Reibold et al., “Carbon Nanotubes in an Ancient Damascus Sabre,” *Nature* 444 (2006)

20.8 Knife Shapes and Their Uses

- Victorinox: Which knife does what
- KAI Europe: Japanese blade families
- KAI 2025-26 kitchen-knife catalog
- Tojiro: Western-style knife categories
- ZWILLING: Knife types and Santoku profile
- Knife Steel Nerds: Why Doesn’t Heat Treating Affect Steel “Flex?”
- Knife Steel Nerds: What a Good Heat Treatment Can and Cannot Do

Source note: Xinzuo product names and examples are concrete case studies in a general technical explanation. Catalogue-derived product photographs remain subject to the image-rights notice.

20.9 Weight, Balance and Ergonomics

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Additional shape references: ZWILLING describes the Santoku’s characteristic high blade and compact profile; specialist comparisons consistently identify the Bunka’s fine K-tip as the more precise and delicate front geometry.

20.10 Learning Hand Sharpening

- *Knife Sharpening Course: Japanese Knives / Western Knives*. Japanese-language workshop reference supplied to the project as a scan with machine-translated OCR; used critically for stone behaviour, contact diagnosis, bevel geometry and single-bevel practice.
- Natural Whetstone and Hone Museum, Kameoka
- Kyoto Prefecture Tourism, Natural Whetstone and Hone Museum